

PHYSICS GRADE 10: PROPERTIES OF WAVES

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.1 (12 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 2.0: Waves and Radioactivity
Sub-Strand	Sub-Strand 2.1: Properties of Waves
Total Duration	12 lessons × 40 minutes = 480 minutes total
Sub-Strand Content	– Rectilinear propagation of waves – Reflection of waves (laws of reflection, ripple tank and light) – Refraction of waves (change in speed and direction, Snell's law idea) – Diffraction of waves (effect of wavelength and gap size) – Interference of waves (constructive and destructive, path difference) – Standing (stationary) waves on strings and in air columns – Doppler effect (moving source and observer) – Applications of wave properties in day-to-day life (radio and FM communication, mobile phone networks, sonar, ultrasound scans)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - describe the rectilinear propagation of waves using ray and wavefront diagrams, - investigate reflection, refraction, diffraction and interference of waves using ripple tanks, strings, light and sound, - explain the formation of standing waves and the Doppler effect using the wave equation $v = f\lambda$, - apply the properties of waves to explain everyday phenomena such as echoes, radio and FM reception, and the changing pitch of passing sirens, - appreciate the application of wave properties in communication, medicine and technology in Kenya.
Core Competencies	– Critical thinking and problem solving: learners analyse wave evidence from simulations and experiments to explain why waves bend around obstacles, cancel out or change pitch. – Communication and collaboration: learners work in groups to run ripple tank and string investigations and present findings and models to the class. – Digital literacy: learners use PhET simulations (Wave Interference, Bending Light, Geometric Optics, Standing Waves on a String), videos and Khan Academy resources to gather evidence. – Learning to learn: learners revisit the Driving Question Board and revise their models as new evidence emerges.
Core Values	– Responsibility: learners take care of apparatus and fulfil assigned roles in group investigations. – Unity: learners cooperate to collect and pool data and build shared class models. – Respect: learners listen to and value the explanations and predictions of peers.

	– Integrity: learners report observations honestly, even when results differ from predictions.
Science & Engineering Practices	– Asking questions and defining problems: learners generate questions about the matatu sound mystery for the Driving Question Board. – Developing and using models: learners draw and revise wavefront, ray and standing-wave models across the unit. – Planning and carrying out investigations: learners use ripple tanks, slinky springs and strings to test reflection, refraction, diffraction and interference. – Analysing and interpreting data: learners analyse frequency, wavelength and speed data using $v = f\lambda$. – Using mathematics and computational thinking: learners calculate wave speed, frequency and wavelength, and simulate waves digitally. – Constructing explanations: learners explain the anchoring phenomenon using evidence. – Engaging in argument from evidence: learners defend or revise predictions in class discussions. – Obtaining, evaluating and communicating information: learners read and present on FM radio, sonar and ultrasound.
Pertinent & Contemporary Issues (PCIs)	– Safety and Security Education: learners handle electrical ripple tank equipment safely and learn about the dangers of loud noise to hearing. – Environmental Education: learners link noise pollution in towns like Nairobi and Mombasa to sound wave behaviour. – Health Education: learners relate ultrasound scans and hearing protection to wave properties. – Digital Citizenship: learners use online simulations and videos responsibly.
Career Connections	– Telecommunications engineer: designs mobile phone and FM radio networks for Safaricom, Airtel and radio stations across Kenya. – Radiographer or sonographer: uses ultrasound waves for medical imaging at hospitals such as Kenyatta National Hospital. – Meteorologist: uses radar and Doppler effect to track weather at the Kenya Meteorological Department. – Audio engineer or acoustician: designs sound systems and halls, for example at the Kenya National Theatre. – Marine and fisheries technologist: uses sonar to locate fish in Lake Victoria and the Indian Ocean. – Physics teacher or researcher: teaches and investigates wave phenomena.
Focus for Lessons	This unit uses one everyday puzzle, hearing a matatu's music around a corner while its pitch changes as it passes, to drive the study of wave properties. Learners predict, observe through simulations and hands-on investigations, explain with the wave model, and keep revising a class model and Driving Question Board across 12 lessons. The unit moves from rectilinear propagation through reflection, refraction, diffraction, interference and standing waves to the Doppler effect and real-world applications such as FM radio.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we hear a matatu around a corner that we cannot see, and why does its pitch change and its radio signal fade as it moves? KICD KEY INQUIRY QUESTIONS: 1. How do waves behave when they meet obstacles or different media? 2. How are the properties of waves applied in day-to-day life?
Anchoring Phenomenon	On a busy Nairobi street, a matatu blasting music is parked behind a building. Students standing around the corner cannot see the matatu, but they can clearly hear the music, even though a torch beam from the matatu would be blocked by the wall. When the matatu drives past, the pitch of its horn and music is higher as it approaches and suddenly lower as it moves away, even though the driver does not change the sound. On the same trip, the FM radio inside the matatu fades and crackles at certain spots near the Ngong Hills and comes in clearly again a few metres on. Students find this puzzling: why do sound waves seem to bend

	around obstacles when light does not, why does the pitch change for a moving source, and why does the signal strength vary from place to place if the radio station never changes?
Storyline Thread	Lesson 1: Introduce the matatu phenomenon; students predict, record initial models and generate questions for the Driving Question Board (DQB). Lesson 2: Rectilinear propagation of waves; use the PhET Geometric Optics simulation and ripple tank wavefronts to show how waves travel in straight lines in a uniform medium. Lesson 3: Reflection of waves; investigate the laws of reflection with ripple tanks, plane mirrors and echoes, and revise the model. Lesson 4: Reflection applications; explain echoes, sonar fish-finding on Lake Victoria and the effect of hall design on sound, and add to the DQB. Lesson 5: Refraction of waves; investigate speed and direction change using ripple tank depth changes, PhET Bending Light and YouTube demonstrations. Lesson 6: Refraction calculations and applications; use $v = f\lambda$ to explain why wavelength changes but frequency does not, and explain the bent straw. Lesson 7: Diffraction of waves; use PhET Wave Interference and the ripple tank to see how gap size and wavelength affect bending, and begin explaining why sound turns the corner but light does not. Lesson 8: Diffraction and the corner puzzle; compare sound and light wavelengths, then revise the model to explain the matatu music heard around the building. Lesson 9: Interference of waves; investigate constructive and destructive interference using two ripple sources and speakers, and explain the loud and quiet spots and fading FM radio. Lesson 10: Standing waves; use the PhET Standing Waves on a String simulation and a slinky or guitar string to relate string length, tension, nodes and antinodes to musical notes. Lesson 11: Doppler effect; use a YouTube demonstration, Khan Academy and a classroom moving-buzzer activity to explain the changing pitch of the matatu horn and the ambulance siren. Lesson 12: Applications and synthesis; explore FM modulation and ultrasound, then finalise the class model, revisit the DQB and explain the whole matatu phenomenon in a written and oral presentation.
Supporting Phenomena	– An echo returns from a rock face at Hell's Gate or a hall wall after you shout, but not in an open field. – A straw looks bent in a glass of water, and ripples slow down and change direction when they pass from deep to shallow water in a basin. – A plucked guitar string or nyatiti string produces a clear musical note only at certain lengths and tensions (standing waves). – An ambulance siren on Mombasa Road sounds higher-pitched when approaching and lower when leaving (Doppler effect). – Two speakers playing the same note in a hall create loud spots and quiet spots (interference).

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (80 minutes (2 x 40-minute periods)): The Matatu Mystery: Launching the Phenomenon

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the matatu phenomenon so that learners observe three puzzles (sound heard around a corner, pitch change of a moving horn, and fading FM radio near the Ngong Hills), record their own initial ideas, and ask testable questions that will drive the whole sub-strand.
Knowledge	A wave is a disturbance that transfers energy from one point to another without transferring matter. Learners observe that sound is heard around a 4.0 m opening while a torch beam is blocked, that a matatu horn of true frequency 400 Hz is heard at 418 Hz while approaching and 383 Hz while receding, and that a 100 MHz FM station fades at certain spots near the Ngong Hills. These values are only observed in this lesson, not calculated.
Skills	Making predictions from prior knowledge of sound and light (NGSS SEP: Asking Questions and Defining Problems). Developing an initial model with drawings and words (NGSS SEP: Developing and Using Models). Observing and recording evidence from a demonstration (NGSS SEP: Planning and Carrying Out Investigations). Turning puzzling observations into testable questions. Communicating ideas in pairs and to the class (Communication and collaboration).
Attitudes	Curiosity about everyday phenomena on Kenyan roads. Respect for peers' diverse ideas and open-mindedness when ideas differ (Respect). Responsibility in contributing roles within a group. Appreciation of the importance of Physics in day-to-day life.
Key Inquiry Question	1. How do waves behave when they meet obstacles or different media? 2. How are the properties of waves applied in day-to-day life?
Purpose in Storyline	This is the opening lesson. It introduces the anchoring phenomenon, surfaces learners' prior ideas, starts the class model with three parts (A: corner/diffraction, B: pitch/Doppler, C: signal fading/interference) and opens the Driving Question Board (DQB). Later lessons return to the DQB questions: Lesson 2 addresses why the torch beam is blocked by the wall, and later lessons progressively revise Models A, B and C.
Safety Notes	Keep the buzzer-on-a-string swing in a clear space with learners at arm's length from the swinger. Keep the volume of speakers and the horn recording at a safe level and not close to ears. During any outdoor corner demonstration, stay on the school compound away from roads and vehicles. Use a torch carefully and never shine it into anyone's eyes.

B. LESSON OVERVIEW

Mr Kamau opens by describing a real trip: Baraka, a matatu driver on the Nairobi-Ngong route, keeps his horn steady at 400 Hz and his radio tuned to one 100 MHz FM station. Wanjiru and Otieno stand around the corner of a building on a busy Nairobi street. They cannot see the parked matatu hiding behind a concrete wall, yet they hear its music clearly, while a torch beam from the matatu would be blocked. When Baraka drives past at a constant 15 m/s (54 km/h), the horn sounds like 418 Hz while approaching and suddenly 383 Hz while moving away, though Baraka never changes it. Near the Ngong Hills, the radio fades and crackles at certain spots and returns a few metres later, though the station never changes. Learners predict individually, observe a demonstration (a buzzer on a string, a slinky pulse, and a recorded matatu horn and music, with a torch and wall comparison), build a first explanation of what a wave is (a disturbance that carries energy), generate questions for the DQB, and draw an initial three-part class model (A, B, C). Nothing is explained fully in this lesson; the goal is productive puzzlement and testable questions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Properties of periodic waves - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Frequency and Pitch of Sound (Flexbook) Source: CK-12  Search ARES for similar readings	Learners hear the story of Baraka's matatu, Wanjiru and Otieno at the corner, and the Ngong Hills radio. Working silently for 4 minutes in their notebooks, each learner writes predictions and reasons for three prompts: (A) Wanjiru and Otieno cannot see the matatu behind the concrete wall. Will they hear the music? Would a torch beam from the matatu reach them? Why? (B) When the matatu drives past with its horn steady, will the pitch they hear stay the same, rise, or fall? (C) The FM radio is tuned to one station but fades near the Ngong Hills. What might cause this? They then do a Think-Pair-Share, comparing predictions with a partner and marking where they agree or disagree.	Read the phenomenon aloud slowly and say: "Before I tell you anything, I want your honest first ideas. There are no wrong predictions today; wrong ideas are useful to us." Use a clear voice and show a sketch of the street, wall, matatu and the two students on the board. Ask: "Who has ever heard a matatu or boda boda they could not see?" Take 3 hands. After the silent writing, say "Turn to your partner and share your reason, not just your answer." Circulate to at least 6 pairs and listen for words such as 'sound bends', 'sound is louder', 'light goes straight', 'frequency', 'signal'. Then cold-call 4 learners (not volunteers) to share one prediction each, using WAIT TIME of 10-15 seconds before calling and saying "Take your time, I will wait." Record predictions on the board in three columns A, B, C without judging them. Ask: "Who thought something different? What was your reasoning?" Do not give any explanation.	Think-Pair-Share with prediction and reasoning (Predict-Observe-Explain). Writing predictions with reasons makes prior ideas visible to the learner and the teacher, and creates a personal stake in the evidence later, so learners notice when evidence confirms or challenges their thinking.	Check that 90% of learners have written a prediction with a reason for all three prompts A, B and C. Look for indicators of prior ideas: learners who say sound 'goes round' versus those who say it 'reflects off other buildings'; learners who say the horn 'changes by itself' or 'gets louder' versus those who link pitch to speed; learners who blame 'weak signal', 'hills' or 'weather' for fading. Tally on the board the number of learners who predict that sound is heard, that the pitch rises/falls/stays the same, and who attribute fading to the hills. Note learners who confuse pitch and loudness for follow-up.
Observe Phase  VIDEO: Properties of periodic waves Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Frequency and Pitch of Sound Practice Source: CK-12  Search ARES	In groups of 4 with assigned roles (Reader, Recorder, Materials manager, Reporter), learners rotate through three short stations of about 6 minutes each, recording observations and drawings in a table of 'What we saw/heard' and 'What puzzles us'. Station 1 (Corner): a speaker or phone plays music from behind a classroom partition or a wall corner (or learners go outside the classroom door to the corner of the building), while a torch is shone from the same hidden spot. Learners record what they hear and what they see.	Before the stations, say: "Today you are scientists, not answer-finders. Your job is to observe carefully and write exactly what you see and hear, without explaining yet." Show how to fill the observation table with one worked example row. At Station 1, ask: "What can Wanjiru and Otieno hear? What can they see? What did the torch do?" Wait 10 seconds before accepting answers. At Station 2, say: "The driver never touched the horn. What do you notice about the pitch? Write 'higher' or 'lower' and the numbers	Structured observation with Notice and Wonder. Separating 'What we saw/heard' from 'What puzzles us' trains learners to distinguish evidence from inference. Using multiple contrasting observations (sound versus light, rising versus falling pitch, strong versus fading signal) from one phenomenon helps learners see patterns that need explanation.	Collect each group's observation table. Success indicators: records that the music is heard but the torch beam is blocked; records both 418 Hz (approaching) and 383 Hz (receding) against the true 400 Hz; records that the radio signal fades at certain spots though the station is fixed; observations are written separately from explanations. At the slinky, listen for learners saying 'the pulse moved but the coils only moved back and forth'. Mark groups that mix explanations into observations for prompts. Aim for at least 8 of

for similar readings	The observed outcome: the music is heard around the corner but the torch light is blocked. Learners note the 4.0 m street opening in the story is similar to the width of a corridor gap. Station 2 (Pitch): learners listen to a recording of the matatu horn (true frequency 400 Hz) passing by; Mr Kamau tells them a phone app shows 418 Hz heard on approach and 383 Hz heard on moving away, while Baraka keeps the horn at 400 Hz. They also swing a small buzzer on a string in a safe circle and listen from the side to hear the pitch rise and fall. They record only what they hear and the values given, with no calculations. Station 3 (Signal): learners listen to a pre-recorded FM radio clip from the 100 MHz station, with the signal fading, crackling, and returning, played as the matatu moves near the Ngong Hills, and they look at a simple diagram showing the hillside and radio tower. They note the fading spots are metres apart (about 1.5 m, given as an observed value only). Then, with a slinky stretched on the floor between two learners, they send a single push-pull pulse and watch what moves along the slinky and what does not.	418 Hz and 383 Hz, and ask yourselves what could be changing." Clarify that pitch is what we hear related to frequency and loudness is amplitude, and that they are not the same. At Station 3, ask: "The station never changes. What did the signal do?" At the slinky, ask: "When the pulse reached the other end, did the coils travel to your partner? What moved?" Circulate and cold-call 2 groups at each station to read one observation aloud. Use the timer, rotating groups every 6 minutes. After rotation, hold a whole-class debrief: "What was the same across all three puzzles?" Wait 15 seconds and take 4 responses. Do not explain diffraction, Doppler or interference; if learners ask, say: "Great question. Put it on a sticky note for our Driving Question Board."		10 groups to have complete tables.
Explain Phase  VIDEO: Sound Properties: Amplitude, period, frequency, wavelength Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners return to their predictions and discuss with their group: Which predictions matched what we observed? Which did not? Using the slinky and the buzzer, they construct a first explanation of 'What is a wave?' They complete the sentence frame: 'A wave is a _____ that carries _____ from _____ to _____ without carrying _____.' Then they connect the idea to the phenomenon: the horn	Say: "Look at your predictions. Put a tick where the evidence agrees and a question mark where you are not sure." Then ask: "When Wanjiru and Otieno hear the matatu music, what is travelling from the matatu to their ears? Is it air, or something else?" Use WAIT TIME of 15 seconds and cold-call 3 learners. Use the slinky to model: push-pull one end and ask, "What moved along the slinky, and	Claim-Evidence-Reasoning (CER) draft with a gallery walk. Learners state a tentative claim for each puzzle, cite evidence from the stations, and give reasoning with a confidence rating. The gallery walk exposes competing ideas and builds a need for more evidence, reinforcing that science explanations are revised as evidence grows.	Listen for and record whether learners can state that a wave transfers energy without transferring matter (indicator: use of the slinky pulse as evidence). Check each group poster for a claim, at least one piece of evidence from the stations, and a reasoning sentence for each of A, B and C. Look for common misconceptions: 'the horn itself changes', 'sound only travels in

 HTML: Wave on a String Source: PhET Interactive Simulations (English)  Search ARES for similar readings	vibrates the air, the vibration travels as a sound wave to Wanjiru and Otieno's ears; the radio signal is also a wave that travels from the transmitter to the matatu radio. Each group writes a 3-sentence 'best idea so far' for each of puzzles A, B and C and marks each with a confidence level (1-3) and one thing they would need to find out. They share in a gallery walk, adding one sticky-note comment (a question or a 'we think the same because...') to two other groups' posters.	what stayed roughly in place?" Guide learners to the idea that a wave is a disturbance carrying energy, not matter, noting that the music is heard but no air is blown from the matatu to Wanjiru. Say: "This is our starting definition. We will test it throughout the unit." Clarify that this lesson only introduces that waves carry energy, and later lessons will examine how they behave. Circulate during group writing and ask each group: "What would you need to see or measure to convince a sceptical friend?" During the gallery walk, set a 6-minute timer and ensure each learner writes at least one sticky note. Cold-call 3 learners to read one note that challenged their thinking. Avoid giving the real explanations of diffraction, Doppler effect, or interference; respond to such ideas with "Interesting, let's park it on the DQB and test it."		straight lines like light', 'the radio station is moving'. Use a quick thumbs check (up, sideways, down) on 'I can say what a wave carries', and note the number of learners with thumbs sideways or down for reteaching in Lesson 2.
Driving Question Board (DQB) Creation  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Radio Waves (Flexbook) Source: CK-12  Search ARES for similar readings	The class gathers around a large chart or wall space headed with the driving question: 'How can we hear a matatu around a corner that we cannot see, and why does its pitch change and its radio signal fade as it moves?' Each learner writes 2 questions on separate sticky notes that start with 'Why', 'How' or 'What would happen if', and that could be tested. In groups they sort questions into three clusters (A: corner/sound and light, B: pitch change, C: radio fading) plus a fourth cluster 'Other / how waves are used'. They star the questions they most want to answer and rewrite at least one vague question (e.g., 'Why is sound strange?') into a testable question (e.g., 'Does a wider gap	Say: "A good scientific question is one we could test with an experiment, a measurement or a model. Let's practise." Model one rewrite on the board, turning 'Why is the pitch weird?' into 'How does the speed of the matatu change the pitch we hear?' Give 4 minutes of silent writing. Then say: "Place your notes under the heading that fits best. If it fits two, make a copy." Cold-call 4 learners to read a question aloud and explain why it belongs in its cluster, with 10 seconds of WAIT TIME beforehand. Ask: "Which question could we test in this classroom with a ripple tank, two speakers, a slinky, a guitar string or a buzzer on a string? Which needs a simulation?" Highlight links to the	Driving Question Board with question clustering. Sorting and rewriting questions into testable forms gives learners ownership of the unit, shows how their curiosity connects to the three parts of the phenomenon, and provides a visible record against which later evidence can be compared.	Check that every learner has contributed at least 2 questions and that at least 60% are testable (they name something that can be changed, measured or compared). Look for questions in all three clusters A, B and C and at least some linking to everyday applications of waves. Record learners who write only yes/no or purely factual questions for support in later lessons. Observe whether learners can justify cluster placement using evidence from the stations (for example, 'the torch was blocked but the music was not').

	change how much sound bends round a corner?'). Examples may include: 'Does a faster matatu make a bigger pitch change?', 'Would a lower pitch bend around the wall more than a high pitch?', 'Does the radio fade at the same spots every time?'	KICD key inquiry questions by asking: "Which of our questions are about waves meeting obstacles or different media? Which are about how waves are used in daily life?" Tell learners the DQB stays up all unit and that questions will be answered and moved to an 'answered' area with evidence. Photograph the board and keep a record. Do not answer the questions now.		
Model Building Phase  VIDEO: Properties of periodic waves - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Radio Waves Practice Source: CK-12  Search ARES for similar readings	Each group draws an initial class-model poster in three labelled parts. Part A (corner): a top-view sketch of the street, the concrete wall, the 4.0 m opening, the hidden matatu, Wanjiru and Otieno, with arrows showing how they think sound and light travel. Part B (pitch): a sketch of the matatu approaching and then moving away at 15 m/s with the horn at 400 Hz, showing what they think happens to the sound between the horn and the listener, and labelled with the observed 418 Hz and 383 Hz. Part C (signal): a sketch of the Ngong Hills, the 100 MHz FM transmitter, and the matatu, showing how they think the signal reaches the radio and why it fades at certain spots about 1.5 m apart. Under each part they write 'What we think is happening' and 'What we still need to find out' and use a different coloured pen for each part. The class then merges group ideas into one class model on large paper, kept visible for revision in each lesson. Finally, learners complete a 3-row summary table: What we observed, What we learned, What we can explain so far.	Say: "Models are tools for thinking. They are meant to be wrong at first and better later. Draw what you think is happening, including things we cannot see, like the sound itself." Give 12 minutes for drawing. Circulate and press for mechanism with prompts such as: "What is travelling from the horn to the ear in your picture?", "Why did you draw the light and the sound differently?", "Where in your model does the pitch change happen?" Do not correct drawings; instead ask "What evidence from the stations supports that?" Cold-call 3 groups to explain one part of their model, with WAIT TIME of 10-15 seconds after each question, and invite two other groups to ask one question. Build the class model on the wall, using one colour for ideas that most groups share and another for ideas that are disputed. Close with: "Today we have a first model and a board full of questions. Next lesson, we start with the torch and the wall: why does the beam stop, and how do we draw how waves travel?" Have learners fill the summary table as an exit ticket.	Initial (progressive) modelling. Drawing a first model with 'what we think' and 'what we need to find out' makes hidden mechanisms visible, allows later lessons to revise a single class model in three parts (A, B, C), and shows learners that explanations improve as evidence grows.	Use the group posters and exit tickets. Indicators of success: each model includes all three parts A, B and C; Part A shows a different path or behaviour for sound and light; Part B uses the values 418 Hz, 383 Hz and 400 Hz correctly as observations and does not blame the horn; Part C includes the transmitter, the hillside and the matatu radio; each part has a 'still need to find out' note; the exit-ticket table separates observations from tentative explanations. Record at least 80% of groups meeting 4 of these 5 indicators; note common initial ideas to compare with later models.

D. TEACHER REFLECTION

Which of the three puzzles (the corner, the pitch change, the fading radio) generated the most curiosity and the most initial ideas, and why? Did learners separate observations from explanations in their tables? Which prior ideas (for example 'sound bends because it is stronger', 'the horn changes', 'the hills block the signal') will need to be revisited in later lessons? Were the questions on the DQB testable enough, and which learners need support in writing them? Did I resist explaining diffraction, Doppler effect and interference, and did I give enough WAIT TIME? What adjustments are needed so that every learner, including those who spoke less, can contribute to the class model in Lesson 2?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Write what you observed about the three puzzles: the music heard around the corner while the torch beam was blocked by the wall, the matatu horn (true frequency 400 Hz) heard at 418 Hz while approaching and 383 Hz while receding at 15 m/s, and the 100 MHz FM radio fading at certain spots near the Ngong Hills though the station never changes.
What did I learn?	Write what you learned today: a wave is a disturbance that carries energy from one place to another without carrying matter (as shown by the slinky pulse and the sound from the matatu horn), and good scientific questions are testable.
How does this explain the phenomenon?	Write what you can explain so far, and what you cannot yet explain: give your best first idea for each of Parts A (corner), B (pitch) and C (fading signal) in your model, and list one question from the Driving Question Board that you want answered next.

LESSON 2 (40 minutes): Waves Travel in Straight Lines: Rays, Wavefronts and the Blocked Torch Beam

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how waves travel in a uniform medium, so they can explain why the torch beam from the matatu is blocked by the wall and what 'straight-line travel' means for light and for water waves. This gives the first piece of evidence for Model Part A (corner/diffraction).
Knowledge	Rectilinear propagation: waves travel in straight lines in a uniform medium. A wavefront is a line joining points in phase. Rays are drawn perpendicular to wavefronts and show the direction of travel. Light in air travels at 3.0×10^8 m/s. In the ripple tank at 10 Hz in deep water, wavelength $\lambda = 3.0$ cm and $v = 0.30$ m/s, which agrees with the wave equation $v = f\lambda$ ($10 \times 0.030 = 0.30$ m/s).
Skills	Predicting and sketching rays; observing wavefronts in a ripple tank and a simulation; using the PhET Geometric Optics simulation to test light travel past an obstacle; applying the wave equation $v = f\lambda$; developing and using models (NGSS SEP 2); analysing and interpreting data (SEP 4); constructing explanations using evidence (SEP 6).
Attitudes	Curiosity and open-mindedness when comparing sound and light; teamwork and respect for diverse ideas (Communication and collaboration); responsibility when handling the ripple tank and lamp; appreciation of how physics explains everyday Nairobi experiences.
Key Inquiry Question	How do waves behave when they meet obstacles or different media? (Focus: how do waves travel when nothing is in the way, so we can later tell what changes at an obstacle?)
Purpose in Storyline	In Lesson 1 learners saw three puzzles and asked why the torch beam is blocked by the wall while the music is heard around the corner. Lesson 2 establishes the baseline: in a uniform medium waves travel in straight lines. Learners can then explain the blocked torch beam and the sharp shadow. This sets up reflection (Lesson 3), refraction (Lessons 5 and 6) and diffraction (Lessons 7 and 8), where the straight-line rule is bent. Lesson 2 does NOT explain why sound turns the corner; that comes later.
Safety Notes	Keep electrical equipment (ripple tank lamp and vibrator) away from spilled water; wipe up spills immediately and put the tank on a stable bench. Do not look directly into the lamp or torch. Hold the torch steady, never shine it at faces. Learners hold the tank edges only when it is switched off. Keep bags off the floor around the tank.

B. LESSON OVERVIEW

Mr Kamau opens by recalling the Lesson 1 question: why does a torch beam from the parked matatu stop at the wall, even though Wanjiru and Otieno can hear the music around the corner? Learners predict and sketch how light and water waves travel. They observe a ripple tank with a 10 Hz vibrator producing straight wavefronts (deep water, $\lambda = 3.0$ cm, $v = 0.30$ m/s) and use the PhET Geometric Optics simulation and a torch behind a card with a slit to see rays travelling in straight lines and casting sharp shadows. In the Explain phase they define wavefront and ray, check $v = f\lambda$ with the ripple data, and explain the sharp shadow behind the wall for light. They add new questions to the DQB (why does sound NOT make a sharp shadow?) and build Model Part A, v1: waves travel in straight lines in a uniform medium, so light is blocked by the wall, but the sound puzzle remains open.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Equation (Flexbook) Source: CK-12  Search ARES for similar readings	Learners sit in groups of four and recall Lesson 1: Wanjiru and Otieno hear Baraka's matatu music around the building, but a torch beam from the matatu would be blocked by the 4.0 m-wide street's concrete wall. Each learner draws on a card (1) the path of a torch beam leaving the matatu and meeting the wall, and (2) the path of water ripples spreading from a point and passing a straight barrier. Beside each sketch they write one sentence beginning 'I predict the waves travel...' and one reason. Groups compare and agree on one group prediction to share.	Display the Lesson 1 puzzles. Say: 'Last lesson we asked why the torch beam is blocked by the wall. Today we find out how light and other waves travel when nothing is in the way. Draw what you think the path looks like.' Give 3 minutes. Walk round and photograph 3 contrasting sketches (straight, curved, spreading) without commenting on correctness. Say: 'Do not rub out; wrong ideas help us.' Cold-call 3 groups to explain their sketch. After each, WAIT TIME 10 seconds, then ask: 'Who has a different path? What made you choose it?' Record predictions on the board under 'Our predictions'.	Predict-Share-Compare (Elicit prior ideas): committing a drawn prediction in public makes learners aware of their current thinking, so later evidence can be tested against it and ideas that change can be seen as progress.	Look for: (a) how many sketches show a straight path versus a curved one; (b) whether learners use words such as 'ray', 'beam' or 'wave'; (c) whether learners assume sound and light behave the same way (note, do not correct). Tally on a clipboard: aim to identify 3 key alternative ideas to revisit in the Explain phase.
Observe Phase  VIDEO: Properties of periodic waves Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Wave Equation Practice Source: CK-12  Search ARES for similar readings	Station 1 (Ripple tank, teacher-led demonstration with learner helpers): A vibrator at 10 Hz makes a straight dipper bar produce plane wavefronts in deep water. Learners view the pattern on the white screen, measure the distance across 5 wavefronts with a ruler on the screen (5 wavelengths = 15 cm, so $\lambda = 3.0$ cm) and record the 10 Hz frequency. They calculate $v = f\lambda = 10 \times 0.030 = 0.30$ m/s. A point dipper is then used to show circular wavefronts, and learners note that the wavefronts spread but each part of a wave still moves outward in a straight line along the radius. Station 2 (PhET Geometric Optics simulation or torch demonstration): Learners shine a torch through a slit in a card toward a screen, then place a wooden block (the 'wall') in the path. They sketch the rays, the lit region and the shadow. In the simulation they place an object and a light source, and test	Set up the ripple tank before class and test the 10 Hz vibrator. Say: 'Watch the wavefronts. What do you notice about their shape and direction?' WAIT TIME 10 seconds before taking answers. Ask a learner helper to measure across 5 wavefronts and say: 'Why do we measure 5 wavelengths instead of 1?' (reduces reading error). Cold-call 4 learners to report the measured λ and the calculated v. At Station 2 say: 'Move the block slowly. Where does the shadow begin and end? Is the edge sharp?' Circulate; ask: 'If the beam bent, where would the shadow edge be?' Ask a group to hold a string taut from the torch to the screen edge to check the straight line. Remind learners of safety rules for water and lamps. Say: 'Record what you SEE, not what you think should happen.'	Structured observation with data table and Analysing and Interpreting Data (SEP 4): recording the same features (path shape, shadow edge, measured λ) for both water and light lets learners spot the common pattern across different waves rather than treating each demonstration as separate.	Check that each learner's table records (a) 10 Hz, $\lambda = 3.0$ cm, $v = 0.30$ m/s; (b) straight-line rays and a sharp shadow for light; (c) the observation that no light enters the shadow region. Observable indicators: learners using a ruler or string to test straightness; correct unit conversion of 3.0 cm to 0.030 m. Note groups that report a 'fuzzy edge' due to a large torch and ask them how to improve the experiment (narrower slit, darker room).

	whether the rays can ever curve around the object. Light travels at 3.0×10^8 m/s, so it covers the 4.0 m street opening in about 1.3×10^{-8} s, which seems instant. Learners record in a table: wave type, what they saw, path shape (straight/curved), shadow sharp or fuzzy.			
Explain Phase  VIDEO: Properties of periodic waves - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Wave Equation (Flexbook) Source: CK-12  Search ARES for similar readings	Learners use their tables to build a class definition of wavefront (a line joining points in phase, such as all the crests) and ray (a line perpendicular to the wavefronts that shows the direction of travel). They label a diagram of the ripple-tank plane waves with 3 wavefronts spaced 3.0 cm apart and rays drawn at right angles. They state the rule 'In a uniform medium, waves travel in straight lines (rectilinear propagation)' and explain what uniform means (same material, same speed throughout, e.g. air of the same temperature, water of the same depth). Then in pairs they draw a ray diagram for the matatu torch beam hitting the wall from Lesson 1: the rays go straight, the wall blocks them, and a shadow forms behind it where Wanjiru and Otieno stand. They write a 3-sentence claim-evidence-reasoning (CER) explanation of why the torch beam is blocked. They end by writing one question: 'If waves travel in straight lines, why can we still hear the music?'	Say: 'Use your data. What do the ripple wavefronts and the torch rays have in common?' WAIT TIME 15 seconds, then ask pairs to share with a partner before cold-calling 4 pairs. Introduce the terms: 'A wavefront joins points that are in step, like all the crests. A ray is an arrow showing direction and always meets the wavefronts at 90 degrees.' Draw one example on the board and have learners correct a deliberately wrong version where a ray is drawn along a wavefront. Prompt: 'Why does the shadow have a sharp edge?' Press for evidence: 'What did you see that shows that?' Present the CER frame: 'Claim: ... Evidence: ... Reasoning: ...'. Then challenge: 'The music reaches Wanjiru and Otieno but the torch beam does not. Does today's rule explain the music?' WAIT TIME 10 seconds. Say clearly: 'Today's rule works for light in a uniform medium. We will test sound and corners in later lessons. Hold that question.' Do not explain diffraction now.	Claim-Evidence-Reasoning (SEP 6) with Developing and Using Models (SEP 2): connecting ripple-tank and torch evidence to a general rule through a labelled wavefront-and-ray diagram. The explicit 'does it explain the music?' prompt creates a productive gap that motivates the next lessons.	Check that CER statements include: Claim (light travels in straight lines so the wall blocks it), Evidence (sharp shadow, straight string line, plane wavefronts in ripple tank), Reasoning (rectilinear propagation in a uniform medium). Indicators of understanding: rays drawn at 90 degrees to wavefronts; correct use of 'uniform medium'. Common errors to listen for: 'light is faster so it can't bend', 'sound is not a wave'. Collect 6 exit notes with the question 'Name one thing a ray tells us.'
Driving Question Board (DQB) Creation  VIDEO: Wave properties Source: Khan Academy (English)	Learners return to the class Driving Question Board with the driving question: 'How can we hear a matatu around a corner that we cannot see, and why does its pitch change and its radio signal fade as it moves?' Groups write new questions on sticky notes, such as	Say: 'Look at the board. Which Lesson 1 question can we now answer with evidence?' WAIT TIME 10 seconds. Invite 2 learners to move the sticky note and add the evidence. Say: 'Write one new question that today's evidence made you wonder about. Put your	Driving Question Board as a Question Tracking Tool: moving questions between 'open' and 'answered' with cited evidence makes learning progress visible and gives learners ownership of the investigation sequence.	Count the number of new questions (aim for at least one per group) and check that they link to evidence from this lesson. Indicators: learners cite the ripple tank or torch shadow when justifying a green tick; questions mention sound, edges or bending

- US curriculum) Search ARES for similar videos EXERCISE: Wave Equation Practice Source: CK-12 Search ARES for similar readings	'Why does sound not make a sharp shadow like light?', 'Do sound waves travel in straight lines too?', 'What happens if the medium changes?', 'What happens at the edge of the wall?'. They post notes under three headings matching the three model parts: A (corner/diffraction), B (pitch/Doppler), C (signal fading/interference). They also mark any Lesson 1 questions now answered (e.g. 'Why is the torch beam blocked?') with a green tick and a note, 'Answered in Lesson 2: light travels in straight lines in a uniform medium.'	name on it.' Cold-call 5 learners to read their questions aloud and ask the class to decide which heading each belongs under. Ask: 'Which question needs a different kind of evidence, such as sound?' Group related questions with a marker. Do not answer unanswered questions; say: 'We will return to these as our evidence grows.' Photograph the board for the next lesson.		rather than only restating the driving question. Note whether any learners cluster questions under the correct model part.
Model Building Phase VIDEO: Properties of periodic waves Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Wave on a String <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English) Search ARES for similar readings	Learners open the class model for the matatu phenomenon, which has three parts. They add the first evidence-based entries to Part A (corner/diffraction), version 1: a diagram showing the matatu hidden behind the building wall, the 4.0 m-wide street opening, Wanjiru and Otieno around the corner, and straight rays from a torch stopped by the wall, with a shadow region behind it. They label 'Light: straight rays, blocked by wall, sharp shadow (rectilinear propagation in uniform air, 3.0×10^8 m/s)' and 'Water waves in ripple tank: plane wavefronts, 10 Hz, $\lambda = 3.0$ cm, $v = 0.30$ m/s'. For sound they draw a dotted question mark path from the matatu to the students and write 'We can hear it. Straight-line rule alone does not explain this yet.' Parts B and C are left as 'Not yet investigated.' Finally each learner completes the summary table: what we observed, what we learned, what we can explain.	Say: 'Use only evidence we have collected. Draw what the straight-line rule predicts for the torch beam, then show where sound does something different.' Provide a large model template on chart paper. Give 5 minutes for groups to draw, then a gallery walk of 3 minutes. Cold-call 3 groups to explain one part of their model. WAIT TIME 10 seconds after each explanation before inviting questions. Ask: 'What does your model explain? What does it leave unexplained?' Press: 'What do you think the sound waves do at the edge of the wall?' Accept ideas as hypotheses and record them on a 'To test' list. Close: 'Next lesson we investigate what happens when waves hit a surface and bounce back.' Collect the summary tables.	Cumulative Model Revision (SEP 2): the class model is built in versions, with each lesson adding evidence-based elements and explicitly marking gaps, so learners see how a scientific model grows and where it still fails.	Check that Part A v1 has straight rays blocked by the wall, a shadow region, correct figures (3.0×10^8 m/s; 10 Hz; $\lambda = 3.0$ cm; $v = 0.30$ m/s; 4.0 m opening) and an honest question mark for sound. Indicators of success: learners distinguish what the model explains (torch beam blocked) from what it does not (music around the corner). Listen for learners using the terms 'ray', 'wavefront' and 'uniform medium' correctly. Review summary tables to identify those who need support with the wave equation.

D. TEACHER REFLECTION

Reflect on the following after the lesson: (1) Did learners separate what they predicted from what they observed in the ripple tank and torch activities? Which prediction was most common, and did learners revise it with evidence? (2) Could learners explain why the torch beam is blocked using the words ray, wavefront and uniform medium? Which learners still thought light 'bends a little' or that sound and light behave identically? (3) How well did the ripple-tank measurement of 5 wavelengths work, and did learners correctly convert 3.0 cm to 0.030 m in $v = f\lambda$? (4) Were the DQB questions linked to evidence, and did any learners already ask about edges or bending that you can build on in Lessons 7 and 8? (5) Did you avoid explaining diffraction too early, and how did learners respond to leaving the sound question open? (6) What adjustments to equipment (darker room, narrower slit, offline simulation) or grouping would improve participation, especially for learners who rarely speak?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In the ripple tank at 10 Hz in deep water we saw straight, evenly spaced plane wavefronts with wavelength 3.0 cm, and we calculated $v = 0.30$ m/s. A torch beam through a slit travelled in a straight line, and a block in its path made a dark shadow with a sharp edge.
What did I learn?	In a uniform medium waves travel in straight lines (rectilinear propagation). A wavefront joins points in phase, and a ray is drawn at 90 degrees to the wavefronts to show direction. Light travels at 3.0×10^8 m/s in air and obeys this rule.
How does this explain the phenomenon?	We can explain why a torch beam from the matatu is blocked by the concrete wall: the light rays travel straight and cannot pass through or around the wall, so a shadow forms where Wanjiru and Otieno stand. We cannot yet explain why the music is heard around the corner, and that question stays on the Driving Question Board.

LESSON 3 (40 minutes (one lesson; adjust to school timetable)): Bouncing Back: Reflection and the Laws of Reflection

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how waves bounce off surfaces. They use plane mirrors and ripple tank barriers to discover that the angle of incidence equals the angle of reflection, and they use this to explain how an echo returns from a wall. This prepares them to explain how the Ngong Hills reflect the FM signal (Part C of the class model).
Knowledge	State the law of reflection: the angle of incidence i equals the angle of reflection r , with angles measured from the normal, and the incident ray, reflected ray and normal lie in the same plane. Describe reflection of ripples at a straight barrier (wavelength 3.0 cm and frequency 10 Hz do not change on reflection). Explain that an echo is a sound wave reflected from a wall, using the echo test: a student 85 m from a wall hears the echo after 0.50 s.
Skills	Predicting, observing and measuring angles with a protractor; drawing ray diagrams with a normal; identifying patterns in data (i vs r); using wavefronts and rays to represent reflection; developing and using models (NGSS SEP: Developing and Using Models); analysing and interpreting data (NGSS SEP); constructing explanations.
Attitudes	Curiosity about how waves behave at boundaries; careful, honest measurement; collaboration in groups and respect for others' ideas; appreciation of how physics explains everyday experiences such as echoes and mirrors in Nairobi.
Key Inquiry Question	1. How do waves behave when they meet obstacles or different media? 2. How are the properties of waves applied in day-to-day life?
Purpose in Storyline	Lesson 1 raised three puzzles about the matatu: sound around the corner, the pitch change, and the fading FM signal. Lesson 2 showed that waves travel in straight lines in a uniform medium, using rays and wavefronts. Lesson 3 asks what happens when a wave hits a surface. It gives learners the law of reflection, which they will need later for the Ngong Hills reflected signal (Lesson 9, Part C), reflection at fixed ends of strings (Lesson 10), and echoes and sonar (Lesson 4). Part A (corner/diffraction) and Part B (Doppler) are not changed in this lesson, but learners record that reflection alone does not explain the bending of sound around a corner.
Safety Notes	Keep water in the ripple tank away from electrical sockets and wipe up spills immediately to prevent slipping. Keep the vibrator and lamp cables off the wet floor. For the echo test, clap in an open school field away from roads, with an adult supervising. Handle mirrors carefully to avoid broken glass, and never shine a torch or reflected sunlight into anyone's eyes.

B. LESSON OVERVIEW

Mr Kamau returns to the Nairobi street scene. Wanjiru and Otieno hear the music, but could the sound be reaching them by bouncing off a wall across the street? Learners predict how ripples and light behave at a barrier, then investigate with a ripple tank (10 Hz, wavelength 3.0 cm), plane mirrors and a ray box or torch. They measure angles of incidence and reflection and discover the law of reflection. They then analyse the echo test (student 85 m from a wall, echo after 0.50 s) as evidence that sound reflects like other waves. Finally they add reflection to the class model and to the Driving Question Board (DQB), noting what reflection does and does not explain about the matatu phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: 06 Laws Of Reflection Source: Seavuria  Search ARES for similar videos  HTML: Reflections in a Plane Mirror (PLIX) Source: CK-12  Search ARES for similar readings	Learners sit in groups of four with the picture of the Nairobi street and the matatu behind the building. Mr Kamau asks: 'Could the music reach Wanjiru and Otieno by bouncing off the building across the street? What happens when a wave hits a wall?' Individually, each learner sketches in their notebook (a) a ripple wavefront hitting a straight barrier at an angle and (b) a torch ray hitting a plane mirror at an angle, drawing where they think the wave goes afterwards. They then compare with a partner and add a one-sentence prediction: 'I think the angle the wave leaves at compared with the angle it arrives at will be _____ because _____.' This links Lesson 2 rays and wavefronts to the Lesson 1 puzzle.	Say: 'In Lesson 2 we learned that waves travel in straight lines. What do you think happens when a straight-moving wave hits a wall? Draw it before we test anything.' Circulate and photograph or note 3 different predictions (e.g. wave goes straight into the wall, bounces back along the same path, bounces off at an equal angle). Give 3 minutes of silent drawing, then WAIT TIME of 10 seconds after asking: 'Who can tell us what angle their wave leaves at?' Cold-call 4 learners from different groups and ask them to hold up sketches; do not say who is correct. Record the competing predictions on the board under a heading 'Our predictions about bouncing waves'. Remind groups that ripples are water waves, light is a wave, and sound is a wave.	Predict-Observe-Explain (POE) with Gallery of Predictions: committing to a prediction in writing creates a mental stake, so surprising or confirming evidence in the next phase is noticed and discussed. Comparing varied predictions surfaces misconceptions (for example, that waves 'stick' to the wall or bounce straight back at any angle).	Check sketches for: (a) a barrier drawn as a line and a wave or ray drawn arriving at an angle; (b) any attempt to show a normal or compare angles. Tally how many of the 4 cold-called learners predict equal angles versus straight-back or other. Look for misconceptions such as 'the wave stops at the wall' or 'sound cannot bounce'. Note learners who label wavefronts or rays correctly from Lesson 2.
Observe Phase  VIDEO: Specular and diffuse reflection Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Law of Reflection (Flexbook) Source: CK-12  Search ARES for similar readings	Station work (rotate every 7 minutes, three stations; groups of four with roles: measurer, recorder, drawer, reporter). STATION 1 Ripple tank: a vibrator set at 10 Hz makes straight ripples of wavelength 3.0 cm in deep water. Learners place a straight metal barrier at an angle to the wavefronts and observe the reflected ripples under the stroboscopic light or lamp image. They sketch incident and reflected wavefronts, measure wavelength before and after reflection with a ruler on the screen (still 3.0 cm), and note the frequency stays 10 Hz. STATION 2 Plane mirror: with a ray box or torch with a slit, a sheet of plain paper, a plane mirror and a protractor, they draw the mirror line, a normal at the point of incidence, shine a ray at 20°, 30°, 40°, 50° and 60° from the normal	Before rotating, say: 'Your job is to collect measurements, not to guess. Measure angles from the normal, not from the mirror surface.' Demonstrate once how to draw a normal at 90° to the mirror. At Station 1 ask: 'What is happening to the wavelength and frequency after the wave hits the barrier? How do you know?' At Station 2 ask: 'Is angle i always bigger than, smaller than, or equal to angle r? Show me evidence from your table.' At Station 3 prompt: 'Why do we multiply 85 by 2?' Use WAIT TIME of 10-15 seconds after each question before taking answers. Cold-call 3 groups at each station for a quick report. Move around to check that learners use the protractor correctly and measure from the normal. If time is short, run Stations 1 and 2 as a teacher	Data table pattern-finding with multiple representations (wavefront sketch, ray diagram, numerical table, real-world echo measurement). Learners look for a consistent relationship between i and r across different representations, building evidence that the same rule applies to water ripples, light and sound. This links to Analysing and Interpreting Data (NGSS SEP).	Observe whether learners (1) measure from the normal and not from the mirror surface; (2) record at least 5 pairs of i and r with values within about 2° to 3° of each other; (3) state that the wavelength stayed 3.0 cm and the frequency stayed 10 Hz after reflection; (4) correctly calculate 170 m and 340 m/s in the echo task. Use a quick checklist during rotation and note groups needing help with protractors.

	and mark the reflected ray with pins or pencil dots. They measure angle i and angle r and fill a table. They also check that the incident ray, reflected ray and normal stay flat on the paper (same plane). STATION 3 Echo evidence: learners examine data from the echo test: a student claps 85 m from a school wall and hears the echo after 0.50 s. They calculate the total distance travelled by the sound ($2 \times 85 = 170$ m) and the speed ($170/0.50 = 340$ m/s), and compare with the speed of sound in air near Nairobi, 340 m/s. They discuss how this shows that sound reflects off a wall.	demonstration and give the echo data as a worked sheet. Keep a timer visible and signal rotations.		
Explain Phase  VIDEO: 06 Laws Of Reflection Source: Seavuria  Search ARES for similar videos  EXERCISE: Law of Reflection Practice Source: CK-12  Search ARES for similar readings	Back in their groups, learners study their data and write a group claim in the format Claim-Evidence-Reasoning (CER): 'When a wave reflects from a surface, the angle of incidence i _____ the angle of reflection r. Evidence: _____. Reasoning: _____. ' They then redraw one ripple tank diagram with wavefronts and rays, labelling the incident ray, reflected ray, normal, angle i and angle r. With Mr Kamau, the class agrees on the law of reflection: the angle of incidence i equals the angle of reflection r, and the incident ray, reflected ray and normal lie in the same plane. Groups then apply the law to two questions: (a) A ray hits a plane mirror at 35° to the normal. What is the angle of reflection, and what is the angle between the incident and reflected rays (70°)? (b) Using the echo test, explain how a sound wave from the student reaches the wall and returns. They also discuss: 'Does reflection explain why Wanjiru and	Say: 'A claim needs evidence from your own table. Point to the numbers.' Ask: 'Why do we measure angles from the normal instead of from the mirror?' WAIT TIME of 10-15 seconds, then cold-call 3 learners. Ask: 'In the ripple tank, the wavelength stayed 3.0 cm and the frequency stayed 10 Hz. What does that tell you about the wave speed using the wave equation $v = f\lambda$?' (Expected: $v = 10 \times 0.030 = 0.30$ m/s, the same as before.) Introduce the formal wording of the law and write it on the board. Address misconceptions: a rough wall still reflects, but scatters in many directions, while a smooth mirror gives one clear reflected ray. Press the connection to the phenomenon: 'If the music bounced off a wall opposite the matatu, would that explain why we hear it around a corner at all places, or only where the reflected path reaches? What other idea do we still need?' Do not teach diffraction or the Doppler effect	Claim-Evidence-Reasoning (CER) with Consensus Discussion: requiring learners to link a claim to measured data and to a scientific principle builds argumentation skills (NGSS SEP: Engaging in Argument from Evidence). The explicit 'what reflection cannot explain' question prevents over-generalising a single idea to the whole phenomenon.	Look for CER statements that include numbers from the table (for example '$i = 40^\circ$, $r = 40^\circ$'); correct labelling of the normal on diagrams; correct answers of 35° and 70° in question (a); and the idea that the echo is sound reflected from the wall with a total path of 170 m in question (b). Listen for statements that reflection alone does not explain the bending of sound around the corner or the pitch change. Note learners who still measure from the mirror surface.

	Otieno hear music around the corner, or the pitch change?' and write one sentence on what reflection can and cannot explain so far.	here; tell learners those puzzles will be investigated in later lessons. Cold-call 4 groups to share CER statements and invite respectful challenge: 'Who can add evidence or disagree politely?'		
Driving Question Board (DQB) Creation  VIDEO: Specular and diffuse reflection Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Reflection of Mechanical Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board with the three puzzles from Lesson 1 (sound around the corner, pitch change, FM fading). Each group writes on sticky notes: (1) one question that has been partly answered by today's lesson, and (2) one new question prompted by reflection. Example questions: 'Could the radio signal reflect off the Ngong Hills?' 'If sound reflects, why can we also hear around corners with no wall opposite?' 'Does an echo change pitch?' They place notes on the board and sort them under Part A (corner), Part B (pitch), and Part C (fading signal). One learner from each group explains where their note is placed and why.	Say: 'Look at the board. Which puzzle does reflection help with, and which does it not solve yet?' Give WAIT TIME of 10-15 seconds before taking responses. Cold-call 3 groups to read and place a note. Highlight questions on Part C: 'Remember the FM signal fading near the Ngong Hills. If the hillside reflects the signal, what two signals might arrive at the same spot?' Do not give the full answer; mark this question with a star for Lesson 9. Point out questions that will be answered in Lessons 4 (echoes and sonar), 7 and 8 (diffraction), 9 (interference) and 11 (Doppler effect) so learners see the road ahead. Move any answered questions to a 'Partly answered' column with a one-line note: 'Sound and other waves reflect with $i = r$.'	Driving Question Board with question sorting: connecting new learning to the anchoring phenomenon keeps purpose visible and shows how each investigation contributes to the final explanation. Sorting questions into Parts A, B and C builds the structure of the class model.	Check that sticky notes are specific and testable (for example 'Does the hillside reflect the FM signal?') rather than vague ('Why waves?'). Note correct placement of questions under Parts A, B and C. Observe whether at least two groups connect reflection to Part C and whether learners recognise reflection does not yet answer Part A or B.
Model Building Phase  VIDEO: Reflection and Diffraction of Mechanical Waves Two-Dimensions - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML:	Learners open the class model of the matatu phenomenon with three parts: A (corner/diffraction), B (pitch/Doppler), and C (signal fading/interference). In groups they add a new component called 'Reflection': a diagram of the street showing the matatu, the building and a wall opposite, with rays from the matatu hitting the opposite wall and reflecting with $i = r$ toward Wanjiru and Otieno, and a second diagram of the Ngong Hills as a reflecting hillside sending a signal back toward the matatu radio. They label incident ray, reflected ray, normal, i and r . Under the	Say: 'Add what we have evidence for. Do not add ideas we have not tested yet.' Ask: 'On your model, where would the reflected FM signal meet the direct signal?' WAIT TIME of 10-15 seconds, then cold-call 3 groups. Prompt: 'The matatu is moving at 15 m/s and the FM radio is tuned to a 100 MHz station that does not change. In your model, what does reflection add to Part C? What is still missing?' (Expected: a second, reflected signal arrives, but we still need superposition and interference to explain fading, which comes in Lesson 9.) Make	Cumulative Model Revision with Gallery Walk: revising a shared model after each investigation shows learners how scientific explanations are built from evidence over time (NGSS SEP: Developing and Using Models). The gallery walk allows peer critique and comparison of diagrams.	Check that models show reflected rays with equal angles measured from the normal, correct labels, and a statement that wavelength (3.0 cm) and frequency (10 Hz) stay unchanged on reflection. Look for accurate limits statements such as 'reflection alone does not explain why the FM fades' or 'we still need to explain the corner and pitch change'. Note groups that incorrectly attribute pitch change or bending around the corner to reflection.

Reflections in a Plane Mirror (PLIX) Source: CK-12 Search ARES for similar readings	diagram they write: 'Waves reflect from surfaces with $i = r$; wavelength and frequency stay the same.' They then complete the sentence stems: 'Reflection helps explain _____ but does not yet explain _____ because _____.' Each group revises their initial Lesson 1 model on a poster or in their notebooks and notes one question for Lesson 4.	sure learners do not claim that reflection explains the Doppler effect or sound bending around corners. Let groups display their models for a 3-minute gallery walk. Close by reading the law of reflection aloud together and previewing Lesson 4: 'How can we use the time an echo takes to find distance?'		
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D. TEACHER REFLECTION

After the lesson, review: Did learners measure angles from the normal rather than from the mirror surface? Which groups found $i = r$ from their own data, and which needed support? How did the echo calculation ($2 \times 85 = 170$ m in 0.50 s gives 340 m/s) help learners accept that sound reflects? Did any learners claim that reflection explains the corner or pitch puzzles, and how did you respond? Were the 10-15 second WAIT TIMES used, and did cold-calling include quieter learners? Consider whether the ripple tank worked well in your school; if not, plan a demonstration or use the PhET sound or wave simulation. Note which DQB questions to carry into Lesson 4 (echoes, sonar) and Lesson 9 (the reflected FM signal near the Ngong Hills).

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed ripples (10 Hz, wavelength 3.0 cm) bouncing off a barrier, light rays bouncing off a plane mirror with the angle of reflection equal to the angle of incidence, and an echo returning after 0.50 s from a wall 85 m away.
What did I learn?	Waves reflect from surfaces according to the law of reflection: the angle of incidence i equals the angle of reflection r , measured from the normal, and the incident ray, reflected ray and normal lie in the same plane. Reflection does not change the frequency, wavelength or speed of the wave in the same medium. An echo is a reflected sound wave.
How does this explain the phenomenon?	Sound from the matatu can reach Wanjiru and Otieno by reflecting from walls, and the Ngong Hills can reflect the FM radio signal, giving a second signal that will later be used to explain the fading spots. Reflection alone does not yet explain how sound bends around the corner (Part A), why the pitch changes (Part B), or why the signal fades (Part C), so we need more evidence in later lessons.

LESSON 4 (40 minutes (one lesson)): Echoes, Sonar and Hall Design: Putting Reflection to Work

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply the law of reflection from Lesson 3 to sound, using the time delay of an echo to calculate distance ($d = vt/2$), and explain how reflected sound shapes echoes, sonar fish-finding on Lake Victoria and hall design. This is evidence for the corner puzzle: reflected sound also reaches listeners, so not all sound at the corner needs to bend.
Knowledge	Learners will explain an echo as sound reflected from a hard surface and heard after a time delay. They will use $d = vt/2$ with $v = 340$ m/s in air (85 m gives a 0.50 s echo) and $v = 1500$ m/s in water (0.040 s gives 30 m). They will show that a separate echo needs at least 0.10 s delay, so a minimum distance of 17 m. They will describe how hard surfaces, soft absorbers and surface shape affect the sound in a hall.
Skills	Planning and carrying out investigations (timing an echo). Using mathematics and computational thinking (rearranging $d = vt/2$). Analysing and interpreting data. Constructing explanations. Developing and using models (adding reflection to Model Part A).
Attitudes	Curiosity about everyday sound, respect for others' reasoning during talk, responsibility in group roles, and appreciation of how Physics helps Kenyan fishers and builders.
Key Inquiry Question	1. How do waves behave when they meet obstacles or different media? 2. How are the properties of waves applied in day-to-day life?
Purpose in Storyline	Lesson 3 gave the law of reflection ($i = r$). Lesson 4 shows reflection at work: echoes, Lake Victoria sonar and halls. Learners then ask what still cannot be explained by reflection alone (why sound reaches people who cannot see the matatu, the pitch change, the fading signal), and these questions go on the Driving Question Board (DQB). Refraction (Lesson 5) follows.
Safety Notes	Do the echo test outdoors, away from traffic, with learners on a marked field. Use clapping or wooden blocks only, never loud horns. Keep learners away from the wall base and from roads. Handle the stopwatch carefully. Keep sound short and moderate to protect hearing.

B. LESSON OVERVIEW

Mr Kamau reminds the class that Wanjiru and Otieno could hear the matatu music around the corner. Some sound may have bounced off walls. Learners predict how a clap bounces off a wall 85 m away and how long the echo takes. They then observe the echo test data (0.50 s at 85 m), a PhET Sound simulation of reflection and delay, and a Lake Victoria sonar scenario (0.040 s, 1500 m/s). They explain with $d = vt/2$, why 17 m is the minimum for a separate echo, and how hard and soft surfaces change a hall. The class adds questions to the DQB and revises Model Part A (corner) by noting that reflection is one way sound reaches around buildings, but cannot explain the pitch (B) and fading signal (C) puzzles.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Reflection and Diffraction of Mechanical	Learners picture the matatu parked behind the building on a Nairobi street. On a prediction sheet they answer individually for 3 minutes: (1) Student claps 85 m	Say: 'Remember Wanjiru and Otieno around the corner. Today we ask: could some of that music have bounced off walls to reach them? Before I teach anything, I	Predict-Observe-Explain (POE) with a confidence rating: committing to a prediction before evidence creates a cognitive hook, and the tally shows the class the	Check prediction sheets for: (a) sketch with a ray hitting the wall and returning, with $i = r$ drawn; (b) a time estimate with a reason; (c) any idea of using time to find

Waves Two-Dimensions - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Reflection of Mechanical Waves (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	from a wall. Will they hear a second sound? Sketch the path of the sound using the law of reflection ($i = r$) from Lesson 3. (2) Predict roughly how long the echo takes (choose: 0.05 s, 0.5 s, 5 s) and explain why. (3) A fisher on Lake Victoria sends a sound pulse down to the lake bed. Predict how the fisher could find the depth. They then compare in pairs and mark their confidence.	want your best prediction.' Show a drawing of the clap, wall and student. Ask: 'What do you already know about echoes from the mountains near Kericho or from a gorge?' Give WAIT TIME of 10-15 seconds, silently, after each question. Cold-call 3 learners for predictions and 2 for reasons without saying right or wrong. Say: 'Interesting, hold on to that idea, we will test it.' Record the three time choices as a tally on the board.	range of ideas. Linking to the law of reflection keeps Lesson 3 in active use.	depth. Note learners who think the echo is a different sound or that sound goes through the wall. Indicator: at least 3 of 5 sampled learners link echo to reflection.
Observe Phase  VIDEO: Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Reflection of Mechanical Waves (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Station 1 (outdoor or video evidence): Learners watch or do an echo test. A student claps 85 m from a school wall while others time the echo with a stopwatch. The class data table shows the echo returns after 0.50 s. They repeat at about 10 m and record that the echo is not heard as a separate sound. Station 2 (PhET: Sound Simulation): in groups of 4 they send sound to a barrier, change the distance, and note how the delay changes. Station 3 (Lake Victoria sonar): a data card from a fish-finder in Kisumu says: the pulse returns after 0.040 s, speed of sound in water = 1500 m/s. Learners record in a three-column table: situation, speed, time delay, and what they notice.	Assign roles: timer, recorder, reporter, safety monitor. Say: 'Look at the numbers, what do you notice about distance and delay?' Circulate and ask each group: 'Why does the time double the distance travelled?' Give WAIT TIME of 10-15 seconds before prompting. Prompt: 'Where does the sound go, and how far does it travel in total?' At the 10 m test ask: 'Why can we not separate the echo from the clap?' Cold-call 4 groups to report one observation each. Remind: use 340 m/s for sound in air and 1500 m/s in water. Keep the safety rules: no shouting near roads.	Data table noticing and wondering: learners record observations before explanation, separating what they see from why. Comparing three contexts (open field, simulation, lake) helps them see the same pattern in different media.	Listen for: 'the echo came after 0.50 s', 'closer wall, shorter delay', 'the sound goes there and back'. Check that tables show 0.50 s at 85 m and 0.040 s for sonar. Indicator: 80% of groups record both times correctly and state that the delay increases with distance.
Explain Phase	Learners use the data to build	Say: 'Talk to your partner: why do	Claim-Evidence-Reasoning (CER)	Check that CER statements

 VIDEO: Reflection and Diffraction of Mechanical Waves Two-Dimensions - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Reflection of Mechanical Waves (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	explanations. Step 1: Total distance travelled = $2d$, so speed = $2d/t$, so $d = vt/2$. Step 2: Check the echo test: $d = 340 \times 0.50 / 2 = 85$ m. Step 3: Sonar on Lake Victoria: $d = 1500 \times 0.040 / 2 = 30$ m. Step 4: The ear needs about 0.10 s to separate two sounds, so the minimum distance for a separate echo = $340 \times 0.10 / 2 = 17$ m. Explain why the 10 m test gave no clear echo. Step 5: Hall design: in a hall, many reflections from hard walls arrive late and blur speech; curtains, carpets and ceiling panels absorb sound, reducing echo, and a curved or angled surface directs reflected sound to the audience. Learners write a claim-evidence-reasoning (CER) statement: 'The Kisumu fisher can find the depth of 30 m because...'. Then they apply to a school assembly hall in Nairobi, suggesting two changes to reduce echo.	we divide by 2?' WAIT TIME 10-15 seconds. Cold-call 3 learners to explain using the words 'there and back'. Model the calculation on the board with units, then have pairs do the sonar calculation while you circulate. Ask: 'What would happen to the delay if the lake were 60 m deep?' (0.080 s, double). Ask: 'Why does a bare concrete hall in Mombasa sound echoey but a hall with curtains does not?' Cold-call 2 pairs. Address the misconception that hard surfaces destroy sound: 'The sound is reflected, not lost.' Connect: 'Reflection can send sound around the street, but does it explain why the matatu music is still heard when we are not in its path? Partly. It does not explain the pitch change or the fading, so we need more ideas.'	with worked example then transfer: learners link numerical evidence to a physical reason (sound travels there and back), then apply the same reasoning to a new context (hall design). This builds transfer of knowledge and mathematical modelling.	include the claim (depth is 30 m), evidence (0.040 s and 1500 m/s) and reasoning ($d = vt/2$ because sound travels to the bottom and back). Indicator: 4 of 5 pairs calculate 30 m with units; at least 3 correctly explain why 17 m is the minimum. Note anyone who forgets to halve and re-teach with a diagram.
Driving Question Board (DQB) Creation  VIDEO: Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Reflection of Mechanical Waves	Learners return to the class DQB with the driving question: 'How can we hear a matatu around a corner that we cannot see, and why does its pitch change and its radio signal fade as it moves?' On sticky notes they write (1) what today's evidence helped answer (e.g. echoes show that sound reflects from walls, so some sound can reach the corner by bouncing) and (2) new questions (e.g. 'Does reflection explain the pitch change?', 'Why does the sound bend into the 4.0 m opening even where there is no wall to bounce off?', 'Can radio waves bounce off the Ngong Hills?'). They group notes under Part A, B, C of the model.	Say: 'Look back at our driving question. What did we learn today that moves us forward, and what is still a mystery?' Give WAIT TIME of 10-15 seconds for writing. Ask each learner to write at least 1 new question. Cold-call 4 learners to read and place notes, and ask the class: 'Which part of the matatu puzzle does this belong to: A, B or C?' Highlight a question about radio waves and the Ngong Hills: 'Hold that idea. A reflected wave may matter for the signal.' Do not answer new questions; mark them as 'to be investigated'.	Driving Question Board with categorisation: learners link new evidence to the big question and see which puzzles are still unresolved. This keeps motivation and gives the lesson sequence a purpose.	Check sticky notes: each learner has one 'we learned' and one 'we wonder'. Indicator: at least 70% of new questions are testable and link to A, B or C; look for questions that distinguish reflection from bending (diffraction) or about the radio signal.

(Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings				
Model Building Phase  VIDEO: Reflection and Diffraction of Mechanical Waves Two-Dimensions - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar videos  HTML: Reflection of Mechanical Waves (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners revise the class model in groups, adding to Part A (corner/diffraction) an arrow diagram showing reflected sound: a ray from the matatu hitting a building wall and bouncing towards the street, with $i = r$. They label 'reflection helps some sound reach us' and add a note: 'Reflection alone cannot explain why sound spreads into the 4.0 m opening behind the wall; we need another idea.' In Part C they add a possible clue: 'radio waves may also reflect, e.g. off the Ngong Hills'. Part B (pitch) gets a question mark: 'Reflection does not change the pitch of the horn'. Groups complete a 'What we know now / What we still need to explain' box and gallery-walk.	Say: 'Use arrows and labels. Show where reflection fits into the matatu story.' Provide large paper and markers. WAIT TIME of 10-15 seconds before groups start so they plan. Circulate and ask: 'Where does the sound go after it hits the wall? What angle?' Cold-call 3 groups to explain their revised Part A. During the gallery walk ask each learner to give one piece of feedback. Close: 'Reflection can send sound back to us, but we still cannot say why we hear the matatu when we are not in line of sight, why the pitch changes, or why the signal fades. Next, refraction.'	Collaborative model revision and gallery walk: learners externalise their thinking as diagrams, mark the limits of the model and receive peer feedback. Showing the gaps keeps the evidence sequence going into Lesson 5.	Check models for: a ray diagram with $i = r$, a statement that echo/reflection is part of the explanation, and explicit gaps for B and C. Indicator: all groups add at least one correct reflection element to Part A and one open question. Exit ticket: 'A sonar pulse returns in 0.020 s in water (1500 m/s). How deep?' (15 m).

D. TEACHER REFLECTION

After the lesson, ask yourself: Did learners remember to halve the total distance, or did they use $d = vt$? Which group explained 'there and back' best, and can they help others next time? Did the 17 m minimum make sense after the 10 m test? Did any learners think reflection explains the whole corner puzzle, and how did the DQB help them see its limits? Were the Kenyan contexts (Lake Victoria sonar, halls) engaging? Note which DQB questions to use in Lessons 5 to 8, especially those about bending of waves, and adapt your stopwatch use if timing was inaccurate.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a clap 85 m from a wall returns as an echo after 0.50 s, that a closer wall gives a delay too short to hear separately, and that a Lake Victoria sonar pulse returned after 0.040 s.
What did I learn?	We learned that sound reflects from hard surfaces following the law of reflection, that $d = vt/2$ because the sound travels there and back, that a separate echo needs at least 17 m in air, and that hard and soft surfaces change how a hall sounds.
How does this explain the phenomenon?	We can explain that the sonar depth is $1500 \times 0.040 / 2 = 30$ m and that reflection can send some matatu sound along the street, but reflection alone does not explain why sound reaches people around the 4.0 m opening, why the horn pitch changes, or why the FM signal fades.

LESSON 5 (40 minutes (double period 80 minutes recommended if the ripple tank is set up live)): Refraction: Speed and Direction Change

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate what happens when waves cross a boundary between two media or depths. They find that a change in wave speed at the boundary causes a change in direction (refraction), and that a wave slowing down bends toward the normal. This gives them the idea of waves changing behaviour at boundaries, which supports the later puzzles about the matatu corner, the pitch and the fading radio.
Knowledge	Learners will be able to: (a) state that refraction is a change of direction of a wave caused by a change of wave speed at a boundary; (b) state that a wave bends toward the normal when it slows down and away from the normal when it speeds up; (c) use ripple tank data (deep water $v = 0.30$ m/s, shallow water $v = 0.20$ m/s, frequency 10 Hz in both) to explain why the wavefronts bunch closer together and change direction in shallow water; (d) describe qualitatively that a light ray in air striking water at 40 degrees bends toward the normal to about 29 degrees.
Skills	Developing and using models (drawing wavefront and ray diagrams with the normal); planning and carrying out investigations (observing the ripple tank and the PhET Bending Light simulation); analysing and interpreting data (comparing speeds and angles); constructing explanations; communication and collaboration in groups of four.
Attitudes	Curiosity about how waves behave at boundaries; open-mindedness when revising predictions after evidence; respect for the ideas of peers (value: Respect); responsibility when sharing roles in group work (value: Responsibility); careful and honest observation and recording of data.
Key Inquiry Question	How do waves behave when they meet obstacles or different media?
Purpose in Storyline	Lessons 2 and 3 gave learners rays, wavefronts, the normal and angles, and the law of reflection. Lesson 5 adds a second thing waves do at a boundary: they can enter the new medium and change speed and direction. This extends the class model (Part A, how waves behave at boundaries, and the idea that wave speed matters). Learners meet the idea that wave behaviour depends on wave speed and wavelength, which prepares for Lesson 6 (calculations with $v = f\lambda$ and $n = \sin i / \sin r$) and later lessons on diffraction, interference and the Doppler effect. Refraction is a different idea from diffraction, and the lesson makes sure the two are not confused: the matatu music turning the corner is NOT explained by refraction.
Safety Notes	Ripple tank: keep the mains lamp and power supply away from water spills; use a dry cloth for the floor; do not touch the lamp or sockets with wet hands. Use shallow trays on stable benches. Glass or perspex blocks should be handled by the edges. Do not look directly into the lamp or torch beam. Charge laptops or phones before the lesson for the offline PhET simulation. Report broken glass or spills to the teacher immediately.

B. LESSON OVERVIEW

Mr Kamau begins by reminding the class of the Nairobi street puzzle: Wanjiru and Otieno hear the matatu music around the building, the horn pitch changes as Baraka drives past at 15 m/s, and the FM radio fades near the Ngong Hills. He explains that today the class studies one more thing waves do: what happens when they cross from one medium or depth into another. Learners first predict what will happen to ripple wavefronts and a torch beam when they cross into shallow water or enter water from air. They then observe a ripple tank with a shallow glass plate (deep $v = 0.30$ m/s, shallow $v = 0.20$ m/s, both at 10 Hz), the PhET Bending Light simulation (air to water, 40 degrees becoming about 29 degrees, observed qualitatively), and a short video of a pencil or coin in water. In the Explain phase learners use a claim-evidence-reasoning approach to state that refraction is a change of direction caused by a change of speed and that a slowing wave bends toward the normal. They add

new questions to the Driving Question Board, and in the Model Building phase they add a refraction entry to the class model, deciding honestly which part of the Nairobi puzzle it does and does not explain.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Refraction in a glass of water Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar readings	(8 minutes) Learners sit in groups of four. Mr Kamau shows a picture of a ripple tank with deep water on one side and a shallow glass plate on the other, and a torch shining at an angle on a clear tank of water (the torch beam hits the water surface at 40 degrees to the normal). Learners recall the Nairobi scene: Wanjiru and Otieno hearing the matatu music around the corner. Individually, in their notebooks, they draw (1) how they think straight ripple wavefronts moving at an angle will look after crossing into the shallow region and (2) the path of the torch beam after it enters the water. They write one sentence beginning 'I think the wave will ... because ...'. Groups compare drawings and agree on a group prediction, marking the normal as a dashed line and labelling angles.	Say: 'Yesterday we used rays, wavefronts and the normal. Today the matatu waves, the ripples and the torch beam meet something new: a boundary. Before I show you anything, what do you think happens when a wave crosses into a different medium?' Hand out prediction sheets. Give WAIT TIME of 10-15 seconds of silent thinking before anyone speaks. Say: 'Draw first, talk second.' Circulate and ask: 'Where did you draw the normal? Does your ray bend toward it or away from it? Why?' Cold-call 3 groups to share predictions and record them on the board under 'Our predictions' without saying who is correct. Say: 'Keep your drawing. We will come back to it with evidence.' Probe with: 'Do you think the wave changes speed? What would that do?'	Predict-Observe-Explain (POE) with individual then group drawing. Committing to a prediction in a drawing makes learners' existing ideas visible, so surprises in the evidence stage create genuine conceptual change. Writing a reason ('because ...') trains learners to link ideas to mechanisms rather than guess.	Look for: (1) learners drawing the normal correctly as a dashed line perpendicular to the boundary (Lesson 3 skill); (2) whether predictions show bending, no bending, or reflection only; (3) whether any learner mentions speed in their reasoning. Tally on a clipboard how many predict 'toward the normal', 'away', 'straight on' and 'no idea'. Note common ideas such as 'light and waves always go straight' or 'the wave gets stopped by the boundary'. Use this tally to decide how much time to spend on the ripple tank.
Observe Phase  VIDEO: Refraction in a glass of water Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Speed (Flexbook) Source: CK-12	(14 minutes) Three short stations or demonstrations, rotated or shown to the whole class. Station 1, Ripple tank: Mr Kamau sets the vibrator at 10 Hz. Learners watch straight waves in deep water (wavelength 3.0 cm, $v = 0.30$ m/s) approach a submerged glass plate at an angle, making shallow water. They observe wavefronts closer together in the shallow region (wavelength 2.0 cm, $v = 0.20$ m/s) and the wave direction turning toward the normal. They also see	Say: 'You are scientists collecting evidence. Watch the wavefronts, not just the ripples.' At the ripple tank ask: 'What do you notice about the spacing of the wavefronts in the shallow region? What is the vibrator frequency in each region?' Wait 10-15 seconds after each question. Cold-call 4 learners during the ripple observation: 'What happened to the wavelength?' 'What happened to the direction?' Use a stroboscope or freeze-frame	Structured evidence gathering using multiple representations (live ripple tank, simulation, video). Seeing the same pattern with water waves and light helps learners notice what stays the same across different waves, which is the start of generalising a rule. Recording data in a table forces attention to the variables: speed, wavelength, frequency and angle.	Observable indicators: (1) learners correctly sketch wavefronts closer together in shallow water; (2) learners state that the frequency stays at 10 Hz; (3) learners read about 29 degrees from the simulation for a 40 degree incident ray and describe it as 'bent toward the normal'; (4) learners record bending away from the normal when going from water to air. Check notebooks for the data table and for the normal drawn on each sketch. Ask 3 learners at random:

Search ARES for similar readings	that the vibrator still taps 10 times each second in both regions. Learners sketch the wavefronts before and after the boundary, drawing the normal and measuring a rough angle on each side with a protractor on the screen or paper. Station 2, PhET Bending Light (offline): Learners send a ray from air into water, set the incident angle to 40 degrees, and see the ray bend toward the normal to about 29 degrees (observed qualitatively, with the protractor tool). They try the ray going from water to air and notice bending away from the normal, and try a 0 degree ray to see no bending. Station 3, Video demonstration: a pencil in a glass of water, and a coin that appears when water is poured into a cup. Learners record observations in a table: Situation, speed before and after, bends toward or away from the normal, wavelength before and after. Link: Wanjiru says, 'So waves can change when they cross something.' Learners record one question about how this connects to the matatu puzzle.	(phone video paused) if available. At the PhET station say: 'Set 40 degrees. Read the refracted angle. Is it bigger or smaller than 40?' Ask: 'What happens if the ray hits the water straight on, at 0 degrees?' Check that learners record the speeds (0.30 m/s and 0.20 m/s) as given data, and that they note light slows in water but do not calculate yet (Lesson 6 does the calculations). Say: 'Draw what you SEE, not what you expected.' Move between groups and ask: 'Does your evidence match your prediction? What changed your mind?' Avoid explaining diffraction here: if a learner asks about the corner, say: 'Great question, put it on the Driving Question Board. Today we are looking at crossing into a new medium, not going around a corner.' Keep safety rules on water and electrical equipment visible.		'Where is the normal on your diagram?' Note any learner who measures angles from the surface instead of from the normal and correct at once.
Explain Phase  VIDEO: Wave properties <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Refraction of Mechanical Waves (Flexbook)	(10 minutes) Groups use the Claim-Evidence-Reasoning (CER) framework on a half-sheet. Claim: 'What does a wave do when it crosses a boundary where its speed changes?' Evidence: they cite at least three observations, for example: deep water $v = 0.30$ m/s, shallow $v = 0.20$ m/s, wavefronts bunched closer (3.0 cm to 2.0 cm), frequency stayed 10 Hz, ripple direction turned toward the normal, light at 40 degrees in air bent to about 29 degrees in water. Reasoning: they explain why a change of speed leads to a	Say: 'Write a claim you can defend with evidence. Number your evidence.' Model one CER example with a different context: a wave moving from smooth tarmac road to a muddy track at Ngong. Run the marching-line demonstration with four learners and say: 'Which end of the line hits the slow patch first? What happens to that end? Which way does the line turn?' Wait 10-15 seconds. Cold-call 4 groups to read their reasoning aloud and ask the class: 'Do you agree? What evidence supports it?' Write the	Claim-Evidence-Reasoning (CER) plus an analogy (the marching line). CER makes learners connect each claim to specific data, strengthening scientific argumentation. The marching-line analogy gives a physical mechanism for why a speed change causes a direction change, so the rule 'slows down, bends toward the normal' is understood rather than memorised. Asking whether refraction explains the corner puzzle trains learners to test an idea against the phenomenon.	Check CER sheets for: (1) a claim that mentions both speed change and direction change; (2) at least three correct pieces of evidence using the given numbers (0.30 m/s, 0.20 m/s, 10 Hz, 3.0 cm, 2.0 cm, 40 degrees, about 29 degrees); (3) reasoning that links the slower side of the wavefront to turning. Use a 3-point check: 3 = all parts correct, 2 = partly correct, 1 = unclear. Listen for the correct statement that frequency stays the same. Exit ticket on a slip: 'A wave goes from deep to shallow water. Does it bend toward or away from

Source: CK-12 Search ARES for similar readings	change of direction by using a wavefront picture: the part of the wavefront that reaches the slower region first slows down while the rest is still fast, so the wavefront swings round and the ray turns toward the normal. A marching-line demonstration helps: four learners link arms and walk from a smooth floor (fast) onto a patch of carpet or sand (slow) at an angle; the line swings. Learners then compare with their predictions and write what they have changed. Finally they answer: 'Does refraction explain why Wanjiru and Otieno hear the music around the building? Why or why not?' They decide that the music turning the corner is a different behaviour (the class will study it later) because here the wave goes into a new medium and changes speed, while at the corner it stays in air.	key statement on the board: 'Refraction is a change of direction caused by a change of wave speed at a boundary. A wave that slows down bends toward the normal.' Ask: 'What would happen if the wave sped up instead?' (bends away). Check for the misconception that the frequency changes: 'The vibrator taps 10 times each second in both regions. So what changed: frequency, wavelength or speed?' Ask: 'Is refraction the reason the music turns the corner? Turn and talk for 1 minute.' Cold-call 2 learners. Say clearly: 'We will solve the corner puzzle in later lessons. Today we have one tool: refraction.' Do not teach the equation $n = \sin i / \sin r$ or the wave equation calculations in this lesson; those belong to Lesson 6.		the normal? Why?' Aim for at least 80% correct; reteach with the marching line if below that.
Driving Question Board (DQB) Creation  VIDEO: Refraction of seismic waves <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Refraction of Mechanical Waves Practice Source: CK-12 Search ARES for similar readings	(4 minutes) Learners return to the class Driving Question Board with the central question: 'How can we hear a matatu around a corner that we cannot see, and why does its pitch change and its radio signal fade as it moves?' Each group writes one new question on a sticky note that came from today's evidence, for example: 'Does the speed of sound change in air, so does sound refract?' 'If waves slow down in shallow water, why does the music still bend around the building?' 'Is the bending of the horn pitch related to speed change?' 'Can the FM signal be bent by the hillside?' They place the notes under the headings Corner (A), Pitch (B) or Signal fading (C), or in a new column 'Waves at boundaries'. They also star earlier questions that today's	Say: 'Look at the board. What did we learn today that helps with our big question? What new question do we have?' Give WAIT TIME of 10-15 seconds before groups write. Cold-call 3 groups to read their sticky notes and ask: 'Which part of the matatu puzzle does your question belong to?' Sort the notes with the class. Point to an earlier question about the corner and ask: 'Does refraction answer this one? What is missing?' Prompt: 'Is there a question that refraction can answer for us, and a question it cannot?' Praise careful questions that refer to evidence. Say: 'Notice we are now sure about one thing: crossing into a new medium changes speed and direction. Which parts of the puzzle are still open?' Make sure the questions about Doppler pitch	Driving Question Board with categorising of questions. Learners generate new questions from evidence and sort them against the phenomenon, so they can see their progress and the gaps in their understanding. This keeps the whole sequence coherent and student-driven.	Check that sticky notes are testable and linked to today's evidence (for example, mention speed, bending, or medium) and not just 'why does the matatu sound change?' Count how many groups can place their question in the correct category (A, B, C or boundaries). Note whether any learner recognises that refraction alone does not explain the music at the corner. Record names of learners who contribute questions, to support quieter learners next time.

	evidence has partly answered or ruled out, for example 'Is the music bent by something changing its speed?'	and interference fading are left for later lessons.		
Model Building Phase VIDEO: Speed of Mechanical Waves - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar videos HTML: Bending Light Source: PhET Interactive Simulations (English) Search ARES for similar readings	(4 minutes) Learners add to the class model, which has three parts: A (corner/diffraction), B (pitch/Doppler) and C (signal fading/interference). In their notebooks they draw a small box labelled 'Waves at a boundary: refraction' under the model. They draw a diagram showing deep water to shallow water with a dashed normal, wavefronts and a ray bending toward the normal, with the values 0.30 m/s and 0.20 m/s and the note 'frequency 10 Hz stays the same'. They add a second diagram of a ray from air into water at 40 degrees bending to about 29 degrees. Beneath, they write: 'Refraction is a change of direction caused by a change of wave speed at a boundary. A wave that slows down bends toward the normal.' They then add a status note on each part of the model: Part A, 'Refraction is not the reason the music turns the corner, we still need to find out why'; Part B, 'no change yet'; Part C, 'no change yet'. They add one sentence on how this tool might be useful later.	Say: 'Our class model has three parts. What do we add today, and where does it go?' Wait 10-15 seconds. Cold-call 3 learners to describe the new box in their own words. Ask: 'What does the model now say about speed and direction? Which part of the Nairobi puzzle does it help with, and which part does it not help with yet?' Insist on honest labelling: 'Do not stretch refraction to explain the corner. A good model says what it cannot do yet.' Provide a sentence frame: 'When a wave slows down at a boundary, it ...'. Collect two model drawings to display on the wall. Preview: 'Next lesson we calculate with $v = f\lambda$ and $n = \sin i / \sin r$ to see exactly how much the wavelength and angle change, including the 40 degrees to 29 degrees for light in water and the 3.0 cm to 2.0 cm in the ripple tank.'	Cumulative model revision with explicit limits. Adding a new element and recording what the element does and does not explain keeps the class model evidence-based, teaches that scientific models grow gradually, and prevents over-generalising a new idea to the whole phenomenon.	Check models for: (1) a labelled diagram with the normal, wavefronts and ray bending toward the normal in the slower region; (2) the correct speeds 0.30 m/s and 0.20 m/s and frequency 10 Hz stated as unchanged; (3) the correct rule statement; (4) an honest note that refraction does not by itself explain the music at the corner, the pitch change or the FM fading. Give each learner a quick rating: complete, partly complete, needs support. Plan a short reteach for learners who draw the ray bending away from the normal in the slower medium.

D. TEACHER REFLECTION

After the lesson ask yourself: (1) Did learners see that the frequency stayed at 10 Hz while the speed and wavelength changed? If not, replay the ripple tank video and ask them to count taps per second. (2) How many learners predicted correctly, and how many changed their ideas because of the evidence? (3) Did any learners confuse refraction with diffraction or reflection? Note examples to address in Lessons 6 and 7. (4) Did every learner measure angles from the normal rather than from the surface? (5) Were all learners able to take part in the group work and the marching-line demonstration, including quieter learners and learners with limited access to devices? (6) Did the offline PhET simulation run smoothly on the available devices, and is a back-up needed? (7) Did the cold-calls and WAIT TIME give enough thinking time? Plan the next lesson by preparing the numbers for calculation: 10 Hz, 3.0 cm to 2.0 cm, $n = 1.33$, 40 degrees to 29 degrees, 2.26×10^8 m/s.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In the ripple tank, waves moved from deep water ($v = 0.30$ m/s, wavelength 3.0 cm) into shallow water ($v = 0.20$ m/s, wavelength 2.0 cm) at 10 Hz, and the wavefronts bunched closer and the direction turned toward the normal. In the PhET simulation a light ray in air at 40 degrees entered water and bent to about 29 degrees, toward the normal; going from water to air it bent away.
What did I learn?	Refraction is a change of direction of a wave caused by a change of wave speed at a boundary. A wave that slows down bends toward the normal; a wave that speeds up bends away from the normal. The frequency stays the same while the speed and wavelength change.
How does this explain the phenomenon?	Refraction describes how waves behave when they enter a new medium or depth, but it does not explain why the matatu music is heard around the building, because the sound stays in the same air. We still need other ideas to explain the corner, the pitch change and the fading radio signal near the Ngong Hills.

LESSON 6 (80 minutes (double lesson)): Refraction Calculations and the Bent Straw: Why Wavelength Changes but Frequency Does Not

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To use the wave equation $v = f\lambda$ and the relation $n = \sin i / \sin r$ to show that when a wave is refracted its frequency stays the same while its speed and wavelength change, and to use this to explain the bent straw. This strengthens the wavelength and $v = f\lambda$ reasoning that Lessons 7 to 9 will need for the matatu puzzles.
Knowledge	Learners will: (1) state that frequency does not change when a wave crosses a boundary, because the source sets the frequency; (2) use $v = f\lambda$ to calculate the wavelength in shallow water (10 Hz, 0.20 m/s gives 2.0 cm, compared with 3.0 cm in deep water at 0.30 m/s); (3) use $n = \sin i / \sin r$ to calculate the refracted angle for light entering water ($n = 1.33$, $i = 40$ degrees gives r of about 29 degrees); (4) calculate the speed of light in water, $3.0 \times 10^8 / 1.33 = 2.26 \times 10^8$ m/s; (5) explain the bent straw as refraction of light leaving water.
Skills	NGSS Science and Engineering Practices: Using Mathematics and Computational Thinking (applying $v = f\lambda$ and $n = \sin i / \sin r$), Analyzing and Interpreting Data (comparing ripple tank measurements), Constructing Explanations (the bent straw), Developing and Using Models (revising Model Part A). Practical skills: measuring angles from the normal, reading a ripple tank, using a calculator for sine values.
Attitudes	Curiosity about everyday optical illusions, precision and honesty with numbers and units, perseverance in multi-step calculations, willingness to revise a prediction when evidence contradicts it, respect for peers' ideas during discussion.
Key Inquiry Question	How do waves behave when they meet obstacles or different media? How are the properties of waves applied in day-to-day life?
Purpose in Storyline	Lesson 5 showed that refraction is a change of direction caused by a change of wave speed at a boundary. Lesson 6 makes that idea quantitative: speed and wavelength change while frequency stays constant, which strengthens the $v = f\lambda$ tool. Lessons 7 and 8 use wavelength to explain why sound diffracts around the 4.0 m opening while light does not; Lesson 9 uses it for FM fading. For the driving question, learners now know that a wave's frequency is set by its source and that wavelength depends on speed. This prepares them to think about the matatu horn, which is at 400 Hz, and its wavelength of 0.85 m.
Safety Notes	Keep the ripple tank away from electrical sockets and wipe up spills straight away to avoid slipping. Use low-voltage lamps and vibrators only. Glass blocks and beakers should be handled with care and carried with two hands. Never look directly at the Sun or a bright torch beam. Use plastic straws and clear plastic cups rather than glass where possible.

B. LESSON OVERVIEW

Mr Kamau opens by reminding the class of the street scene: Wanjiru and Otieno hear the matatu music around the corner, the pitch seems to change, and the FM station fades. He tells them they now need a reliable tool for linking speed, frequency and wavelength. In Lesson 5 they saw that waves bend when their speed changes. Today they ask: by how much, and what stays the same? Learners first predict what happens to frequency, speed and wavelength when ripples go from deep to shallow water and what a straw in water will look like. They then observe the ripple tank (10 Hz in both depths), measure the wavelengths of 3.0 cm and 2.0 cm, and carry out a ray-box and glass-block activity for light going from air into water. In the Explain phase they calculate wavelength with $v = f\lambda$, the refracted angle with $n = \sin i / \sin r$, and the speed of light in water of 2.26×10^8 m/s, and then explain the bent straw in a cup of water. They add questions to the Driving Question Board and revise Model Part A (corner/diffraction) by adding a note that wavelength depends on speed and frequency is fixed by the source. The lesson does not yet explain diffraction at the corner (Lessons 7 and 8), interference (Lesson 9) or the Doppler effect (Lesson 11).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Refraction and Snell's law Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Equation (Flexbook) Source: CK-12  Search ARES for similar readings	(About 12 minutes) Learners sit in groups of four. Mr Kamau shows a plastic straw standing in a clear cup of water, viewed from the side, and recalls the matatu scene on a busy Nairobi street. Each learner privately writes predictions on a Predict Sheet: (1) When ripples made by a 10 Hz vibrator move from deep water into shallow water, where they travel more slowly (0.30 m/s in deep water, 0.20 m/s in shallow water), what happens to the frequency? The wavelength? (2) Why does a straw in water look broken at the surface? (3) Is the straw really bent? They draw a quick sketch to show what they think a ray of light does as it passes from air into water. Groups then compare predictions and agree on one group claim on a mini-whiteboard. Learners link this to the matatu puzzle by writing one sentence: 'If waves change speed, what might change about them that could matter for a sound turning a corner?'	Hold up the straw and say: 'Look carefully. Is this straw really broken? Write your own idea first. Do not talk yet.' Give WAIT TIME of 10-15 seconds, silently counting, before taking any answers. Then say: 'Last lesson we saw that waves bend when their speed changes. Today we ask by how much, and what stays the same.' Circulate and record misconceptions you hear, such as 'the frequency gets smaller in shallow water' or 'the straw really bends in water'. Cold-call 4 groups to read their claims without evaluating them: 'Group 2, what did you decide about the frequency, and what makes you say so?' Write the claims on the board under 'Our predictions'. Say: 'Keep these. We will test them with evidence today.'	Predict-Observe-Explain (POE) with a private write-first routine. Committing to a prediction in writing before the evidence creates cognitive conflict if the prediction is wrong. It makes learners' existing ideas visible so they can later be compared with data. Group claims on whiteboards encourage learners to argue from reasoning, not from authority.	Look for: (a) the number of learners who write that frequency stays the same, decreases, or increases, and tally the responses on the board; (b) whether sketches show light bending at the water surface or the straw physically bending; (c) whether learners use words from Lesson 5 such as speed, normal or bending; (d) whether the whiteboard claims include a reason. Indicator of readiness: at least half of the groups give a reason that mentions speed. Note common misconceptions for follow-up in the Explain phase.
Observe Phase  VIDEO: Properties of periodic waves Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Wave Equation Practice Source: CK-12	(About 25 minutes) Learners rotate through two stations in their groups, about 10 minutes each, then share data. Station A: Ripple tank. Using the vibrator at 10 Hz, learners watch the plane waves cross from deep water into a shallow region (a glass plate under the water). They use a stroboscope or photo of the pattern with a ruler to measure the wavelength in deep water (3.0 cm) and in shallow water (2.0 cm), count the number of waves	Before the rotation say: 'At each station your job is to collect numbers. Measure angles from the normal, not from the surface.' Circulate and ask each group at Station A: 'What is the frequency in deep water? In shallow water? How do you know?' and at Station B: 'What do you notice about $\sin i$ divided by $\sin r$ as the angle changes?' Give WAIT TIME of 10-15 seconds after each question. Ask 3 groups to share their wavelength data and 3 to share	Data Collection and Pattern Noticing with a class data chart. Producing the same table of results as a class lets learners see consistent patterns (frequency unchanged, wavelength and speed reduced, $\sin i / \sin r$ constant) and builds confidence that the pattern is real and not an error by one group. Delaying explanation until after the data are shared keeps the focus on evidence.	Look for: (a) learners measuring angles from the normal correctly; (b) groups reporting frequency of 10 Hz in both depths, wavelengths of 3.0 cm and 2.0 cm, and r of about 29 degrees for i of 40 degrees; (c) learners spotting that $\sin i / \sin r$ stays about 1.33 for different angles; (d) accuracy of table entries. Indicator: each group can say, without prompting, which quantity did not change at the boundary. Use a checklist while circulating, and note groups

Search ARES for similar readings	passing a point in 10 s to check the frequency (about 100 waves in 10 s, 10 Hz in both), and note that the waves bend toward the normal in the shallow region. They record: frequency 10 Hz in both depths, wavelength 3.0 cm and 2.0 cm, speed 0.30 m/s and 0.20 m/s from the teacher's timing. Station B: Light in water. Learners use a ray box, a semicircular transparent tank or glass block, and a protractor to send a ray into the water at an angle of incidence of 40 degrees, measuring the refracted angle of about 29 degrees from the normal. They record i and r in a table for 20, 30, 40, 50 and 60 degrees and calculate $\sin i$ and $\sin r$. They then view the straw in the cup from different sides and sketch what they see. Data are entered on a class chart.	their angle data, cold-calling by group number. Prompt: 'Which quantity stayed the same across the boundary? Which two changed?' Say: 'I want you to notice a pattern, not yet explain it.' Ensure safety by keeping water away from sockets and reminding learners to wipe spills. Collect all numbers on a class data chart under headings: Quantity / Deep water / Shallow water.		needing help with the protractor or with counting waves.
Explain Phase VIDEO: Properties of periodic waves - dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos HTML: Refraction of Mechanical Waves (Flexbook) Source: CK-12 Search ARES for similar readings	(About 25 minutes) Using the class data, learners build the explanation step by step. Step 1: Frequency. Learners discuss why the frequency stays 10 Hz: the vibrator sets how many waves are made each second, and the same number of waves per second must cross the boundary, otherwise waves would pile up or vanish at the boundary. Step 2: Wavelength. Using the wave equation $v = f\lambda$, they calculate: deep water $\lambda = v/f = 0.30/10 = 0.030$ m = 3.0 cm; shallow water $\lambda = 0.20/10 = 0.020$ m = 2.0 cm, matching what they measured. Step 3: Refractive index. They use $n = \sin i / \sin r$ to find the refracted angle: $\sin 40$ degrees = 0.643; $\sin r = 0.643/1.33 = 0.483$; $r =$ about 29 degrees, matching Station B. Step 4: Speed of light in water: $n = c/v$,	Say: 'We have numbers. Now let us make sense of them. First, what stayed the same and why?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners: 'Tell me in your own words why the frequency cannot change at the boundary.' If answers are vague, ask: 'If 10 waves leave the vibrator each second, how many must cross the boundary each second?' Model the first calculation on the board with units, then let learners do the second themselves in pairs while you circulate. Ask: 'Using $v = f\lambda$, if f is fixed and v falls, what must happen to λ?' Cold-call 2 pairs to show working. For the bent straw say: 'Draw the ray from the straw tip to your eye. Which way does it bend leaving the water and why?' Have 2 learners come to the board to draw the diagram	Claim-Evidence-Reasoning (CER) combined with worked-example modelling. Learners state a claim (frequency is constant, wavelength changes), cite evidence from the class data chart, and give reasoning with $v = f\lambda$ and $n = \sin i / \sin r$. Linking calculation to measured values shows that mathematics describes real behaviour and strengthens the connection between the equation and what they observed. Drawing the ray diagram makes the invisible path of light visible.	Look for: (a) correct use of $v = f\lambda$ with units: 3.0 cm and 2.0 cm; (b) correct calculation of r of about 29 degrees and v of 2.26×10^8 m/s; (c) a written explanation of the bent straw that states light bends as it leaves water, the eye traces rays straight back, and the straw itself is straight; (d) a diagram with the normal and angles measured from it. Indicator of mastery: learners can answer 'Why does wavelength change but frequency not?' using the source and speed. Use a 3-question exit check (frequency, wavelength, straw) and sort responses into secure, partly secure and not yet secure to plan the next lesson's starter.

	so $v = 3.0 \times 10^8 / 1.33 = 2.26 \times 10^8$ m/s, slower than in air. Step 5: The bent straw. Light from the part of the straw under water bends away from the normal as it leaves the water into air, so the eye traces the rays back in straight lines and sees the straw's underwater part at a shallower apparent position, so the straw looks broken or bent. The straw itself is not bent. Learners write a four-sentence explanation with a labelled ray diagram showing the normal, the bent ray and the apparent position. They then complete a short practice problem set: a Kisumu fisher on Lake Victoria looks at a fish and misjudges its position because of refraction, and a wave with $f = 5$ Hz and $v = 0.25$ m/s that slows to 0.15 m/s has wavelength 5.0 cm and then 3.0 cm.	and invite the class to critique it. Address the misconception that the straw really bends by asking: 'How could we check that the straw is straight?' (lift it out). Remind the class that diffraction at the corner is for later lessons; today is only refraction.		
Driving Question Board (DQB) Creation  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Refraction of Mechanical Waves Practice Source: CK-12  Search ARES for similar readings	(About 8 minutes) Learners revisit the class Driving Question Board, headed 'How can we hear a matatu around a corner that we cannot see, and why does its pitch change and its radio signal fade as it moves?' In groups they write on sticky notes: (1) a question that has now been answered by today's evidence (for example 'Does the frequency of a wave change when it enters a new medium?'), and (2) a new question prompted by today's work (for example 'If wavelength changes with speed, what does wavelength have to do with sound bending around the 4.0 m opening?' or 'Why does the matatu horn's frequency seem to change when its source frequency is steady?'). They place the notes on the DQB under the three headings Corner (A), Pitch (B) and Signal fading	Say: 'Which of our questions did today's evidence help to answer? Which new questions have appeared?' Give WAIT TIME of 10-15 seconds before groups write. Cold-call 4 groups to read one note aloud and explain where to place it. Ask: 'Does this note belong with Corner, Pitch, or Signal fading?' Point out that the wavelength question links to Part A, to be investigated in Lessons 7 and 8. Praise precise questions and help refine vague ones: 'Can you make that question testable?' Do not answer the new questions yet; say, 'We will come back to these in the coming lessons.' Photograph the board.	Driving Question Board with student-generated questions. The board keeps learning visible and purposeful by showing how each lesson's evidence moves the class closer to the driving question. Sorting questions into the three puzzles builds a coherent storyline and gives learners ownership of what to investigate next.	Look for: (a) questions that are specific and connected to evidence from today; (b) correct sorting of notes into Corner, Pitch or Signal fading; (c) at least one new question per group linking wavelength to the phenomenon; (d) learners moving answered questions to the 'Evidence so far' area with a lesson reference. Indicator: at least three groups produce a question that links wavelength to bending around the corner, showing they are connecting refraction to the next steps.

	(C), and move answered questions to an 'Evidence so far' area with a note of the lesson number.			
Model Building Phase  VIDEO: Refraction and Snell's law Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Equation (Flexbook) Source: CK-12  Search ARES for similar readings	(About 10 minutes) Learners return to the class model, which has three parts: A (corner/diffraction), B (pitch/Doppler) and C (signal fading/interference). In groups they add today's evidence to the model poster. Under a new 'Tools we have' box they write: 'The wave equation $v = f\lambda$: the source fixes f ; when v changes, λ changes.' and ' $n = \sin i / \sin r$; $n = c/v$.' They add a data note: 10 Hz, 3.0 cm to 2.0 cm, $n = 1.33$, 40 degrees to 29 degrees, 2.26×10^8 m/s. For Part A they add: 'Wavelength depends on speed and frequency: for the matatu horn at 400 Hz in air, $\lambda = 340/400 = 0.85$ m.' They mark with a question mark what is still unknown: why wavelength compared with the 4.0 m opening decides whether waves bend around a corner. Parts B and C are left unchanged except for a note that the tool $v = f\lambda$ will be needed. Each group draws a small labelled diagram of the bent straw on the model as an example of refraction. Finally, each learner completes the summary table prompt in their notebook.	Say: 'Our model must always match our evidence. What can we now add and what must we still leave as a question?' Give WAIT TIME of 10-15 seconds. Cold-call 3 groups to state one addition and its supporting evidence. Challenge: 'Does refraction explain why sound turns the corner? How do you know?' Guide learners to see that it does not yet, because the corner involves a gap and an obstacle, not a change of medium. Keep Parts B and C honest: no Doppler or interference explanation is added yet. Ask learners to use words from Lessons 5 and 6 only. Collect the summary tables as an exit ticket and sample three for feedback before the next lesson.	Consensus Model Building and revision. Learners add only claims supported by evidence and mark open questions, so the model grows cumulatively and honestly. Linking a mathematical tool to the phenomenon ($\lambda = 340/400 = 0.85$ m) shows how new knowledge becomes part of a larger explanation. Marking what remains unknown sets up purposeful investigations in Lesson 7.	Look for: (a) model additions that cite specific evidence (10 Hz, 3.0 cm to 2.0 cm, $n = 1.33$); (b) correct statement that frequency is constant and wavelength changes with speed; (c) honest question marks for the unexplained parts; (d) the bent straw diagram correctly labelled. Indicator: groups do not wrongly claim that refraction explains the corner effect. Use the summary table exit tickets to identify learners needing a quick revision of $v = f\lambda$ at the start of Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect on these questions: (1) Did learners explain the constant frequency using the idea of the source setting it, or only repeat the rule? Which questions got the best responses? (2) Which learners could use $v = f\lambda$ and $n = \sin i / \sin r$ correctly and which struggled with sine values or units (cm and m)? (3) Did any learners still think the straw physically bends, and what evidence changed their minds? (4) Was the ripple tank data clear enough, or do the wave counts need a stroboscope or slow-motion video next time? (5) Were all learners able to take part in the cold-calls and DQB tasks? (6) How well do the learners' new questions prepare them to link wavelength with the 4.0 m opening in Lessons 7 and 8? Plan a brief starter for Lesson 7 that revisits wavelength and the wave equation for learners who were not yet secure.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In the ripple tank at 10 Hz, the waves had a wavelength of 3.0 cm in deep water and 2.0 cm in shallow water and bent at the boundary. A ray of light entering water at 40 degrees bent to about 29 degrees from the normal. A straw in water looked broken at the surface.
What did I learn?	When a wave is refracted, its frequency stays the same because the source sets it, while its speed and wavelength change ($v = f\lambda$). The refractive index $n = \sin i / \sin r = c/v$; for water $n = 1.33$ and light travels at 2.26×10^8 m/s. Wavelength depends on the speed of the wave and the frequency of the source.
How does this explain the phenomenon?	The straw looks bent because light from its underwater part bends away from the normal as it leaves the water, and our eyes trace the rays back in straight lines to a shallower apparent position. The straw itself is straight. This does not yet explain why sound turns the corner, but we now have the tool $v = f\lambda$ to work out wavelengths, such as 0.85 m for the matatu horn at 400 Hz, which we will use in Lessons 7 and 8.

LESSON 7 (80 minutes (2 x 40-minute periods)): Diffraction: Gap Size and Wavelength

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how waves spread through gaps and around edges, and discover that diffraction is strongest when the wavelength is comparable to or larger than the gap. They meet the 4.0 m street opening as the real gap in the matatu phenomenon, ready to be tested with sound and light in Lesson 8.
Knowledge	Diffraction is the spreading of waves around an obstacle or through a gap. It is greatest when the wavelength is comparable to or bigger than the gap. Frequency and wavelength are linked by the wave equation $v = f\lambda$ (Lesson 6). In the ripple tank at 10 Hz, deep water has $v = 0.30$ m/s and $\lambda = 3.0$ cm. When the gap is much wider than λ , waves mostly keep travelling straight (Lesson 2), with only slight bending at the edges.
Skills	Planning and carrying out investigations (ripple tank and PhET Wave Interference); measuring and recording gap widths and wavelengths; analysing data to find a pattern (ratio of λ to gap width); using models to predict; constructing an explanation from evidence; communicating findings in diagrams using wavefronts and rays.
Attitudes	Curiosity about why waves behave differently in different situations; patience and precision in observing; respect for peers' ideas during discussion; teamwork and responsibility when sharing equipment; willingness to revise predictions based on evidence.
Key Inquiry Question	1. How do waves behave when they meet obstacles or different media? 2. How are the properties of waves applied in day-to-day life?
Purpose in Storyline	Lessons 1 and 2 left learners asking why the torch beam is blocked by the wall while the music is still heard around the corner. Lesson 7 gives the first piece of evidence: waves spread through gaps, and the amount of spreading depends on the ratio of wavelength to gap width. The street opening is 4.0 m wide and is introduced here as the real gap. In Lesson 8 learners will compare sound and light wavelengths with this opening to fully explain the corner puzzle. This lesson supplies Model Part A (corner/diffraction) with its rule.
Safety Notes	Keep water away from electrical sockets and the vibrator motor; place the ripple tank on a stable table and wipe up spills at once to prevent slipping. Mr Kamau switches the power on and off. Learners must not look directly into the lamp. Barriers should be handled carefully. Take care with any sound demonstration so that loudness stays at a comfortable level.

B. LESSON OVERVIEW

Mr Kamau opens with Wanjiru and Otieno's puzzle: they stand around the corner of a building on a busy Nairobi street and hear the matatu music, yet cannot see the matatu, and a torch beam would be blocked. Learners first predict what happens to ripples at 10 Hz ($\lambda = 3.0$ cm) when they pass through a wide gap and a narrow gap. They then observe the ripple tank with barriers set at different gap widths, and test the same ideas with PhET Wave Interference, recording the ratio of wavelength to gap width. In the Explain phase they build the rule that diffraction is strongest when the wavelength is comparable to or larger than the gap, and are introduced to the 4.0 m street opening as the real gap for the matatu. They add new questions to the Driving Question Board and revise Model Part A, noting that they still need the wavelengths of sound and light to complete the explanation in Lesson 8.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Wave interference Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Interference Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners recall the Lesson 1 puzzle: Wanjiru and Otieno hear the matatu music around the corner but cannot see the vehicle, and the torch beam would be blocked by the concrete wall. Mr Kamau shows the ripple tank with straight waves (10 Hz, $\lambda = 3.0$ cm in deep water) heading towards a barrier with a gap. In pairs, learners draw on a worksheet what they think the wavefronts will look like after passing through (a) a wide gap of 12 cm and (b) a narrow gap of 3 cm. They also write one sentence on which gap will make the waves spread more and why, linking it to the corner behind the building.	Say: "Remember Wanjiru and Otieno standing around the corner of the building. Sound reaches them but the torch beam does not. Today we test what waves do when they meet a gap." Display the barrier set-up but do NOT run the waves yet. Say: "Draw what you think the waves look like on the other side. Use wavefront lines, not arrows." Give 3 minutes of silent drawing, then WAIT TIME of 10-15 seconds after asking: "Which gap do you think spreads the waves more, and why?" Cold-call 4 pairs (two for each prediction) and record predictions on the board without judging them. Ask: "Who has a different drawing from your neighbour? What is different?"	Predict-and-Commit with Drawn Models: learners commit a diagram to paper before seeing evidence. This surfaces the common misconception that waves always travel in perfectly straight lines (an over-generalisation from Lesson 2) and creates a cognitive 'need to know' that the later observation resolves.	Check worksheets for: (1) wavefronts drawn as lines joined across the gap rather than rays only; (2) whether learners show any spreading beyond the gap edges; (3) whether they link gap size to amount of spreading. Tally on the board how many pairs predict 'narrow spreads more', 'wide spreads more', or 'no spreading at all'. Indicator of readiness: at least half the class can justify their prediction with a reason, even if the prediction is wrong.
Observe Phase  VIDEO: Reflection and Diffraction of Mechanical Waves Two-Dimensions - Overview Source: CK-12  Search ARES for similar videos  HTML: Wave Interference Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Station 1 (ripple tank): in groups of four, learners use the 10 Hz vibrator, where deep water gives $v = 0.30$ m/s and $\lambda = 3.0$ cm. Using two barriers they set gap widths of 12 cm, 6 cm, 3 cm and 1.5 cm in turn, and sketch the wavefronts after the gap. They measure the gap width, note $\lambda = 3.0$ cm, and calculate the ratio $\lambda \div \text{gap}$ for each, filling in a table, with a column describing the amount of spreading (very little, some, a lot, almost circular). They also place a single barrier edge in the water and observe whether the waves bend into the 'shadow' behind it. Station 2 (PhET Wave Interference): learners set the slit width and the wavelength and watch waves pass through; they change wavelength while keeping the gap fixed, then change the gap while keeping wavelength fixed. Finally, the class is shown a scale diagram of the street: a concrete	Model the set-up in 2 minutes, then say: "Change only ONE thing at a time. What stays the same? What changes?" Circulate and ask each group: "What is the wavelength here? What is the gap? Which is bigger?" and "Where on the screen do the waves reach beyond the edges of the gap?" Prompt struggling groups: "Look at the edge of the barrier. Do you see waves entering the shadow?" WAIT TIME of 10-15 seconds after each question before offering hints. Pause the class twice to cold-call 3 groups each to report one observation. For PhET, ask: "If you double the wavelength but keep the same gap, what happens to the bending?" Then reveal the street diagram: "This opening is 4.0 m wide. Hold that number in your head; we will use it next lesson." Reinforce safety: only Mr Kamau switches the power.	Structured Data Table and Controlled-Variable Comparison: learners vary one factor at a time and record the ratio $\lambda \div \text{gap}$, which turns observations into a quantitative pattern. Using two different tools (ripple tank and simulation) lets learners check that the pattern holds across systems, which strengthens the claim.	Observe whether groups: (1) change only one variable at a time; (2) measure gap widths with a ruler and record units correctly (cm); (3) calculate ratios correctly (for example, $3.0 \div 12 = 0.25$, $3.0 \div 6 = 0.5$, $3.0 \div 3 = 1$, $3.0 \div 1.5 = 2$); (4) describe the 1.5 cm and 3 cm gaps as giving noticeably circular spreading, the 12 cm gap as mostly straight with slightly curved edges. Checklist entries: accurate table, labelled sketches, and a written statement of the pattern.

	wall with a 4.0 m opening, with Wanjiru and Otieno standing around the corner. Learners note that this is the 'real gap' that sound or light must go through or around.			
Explain Phase  VIDEO: Wave interference Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Interference (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners return to their predictions and compare them with the ripple tank results. In groups they complete a Claim-Evidence-Reasoning (CER) statement: Claim: how does gap width affect the spreading of waves of fixed wavelength? Evidence: data from the ratio table (for example, $\lambda = 3.0$ cm and a gap of 12 cm gave very little spreading, while a gap of 3 cm gave strong, almost circular spreading). Reasoning: using the idea that each point in the gap acts as a source of new waves and that a narrow gap leaves only a small 'source' so the waves spread out, a wide gap behaves like a straight wavefront with only the edges bending. They state the rule in their own words: diffraction is the spreading of waves around an obstacle or through a gap, and it is greatest when the wavelength is comparable to or bigger than the gap. They use $v = f\lambda$ (Lesson 6) to confirm that at 10 Hz and $v = 0.30$ m/s the wavelength is $0.30 \div 10 = 0.03$ m = 3.0 cm, and explain why changing frequency would change the wavelength and therefore the bending. Finally, they discuss a preliminary link to the corner: the opening is 4.0 m; a wave of wavelength much smaller than 4.0 m would behave like the wide-gap case, while a wave with λ closer to 4.0 m would spread like the narrow-gap case. They do not yet compute for sound or light.	Say: "Let's compare your drawing from the start of the lesson with what we saw. Which prediction was closest?" Ask: "What pattern do you see in the ratio column as the spreading increases?" WAIT TIME of 10-15 seconds, then cold-call 5 learners from different groups to state the pattern in one sentence. Push for precise vocabulary: "Use the words 'wavelength', 'gap' and 'diffraction' in your answer." Press on reasoning: "Why does the 12 cm gap hardly spread the waves when the 3 cm gap spreads them a lot?" Check the maths: "Using $v = f\lambda$, what is the wavelength at 10 Hz and 0.30 m/s?" Address the misconception that waves 'only go straight' by referring to Lesson 2: "In Lesson 2 we saw straight-line travel in open space; today we see what happens at edges and gaps." Close with: "The street opening is 4.0 m. Is that a wide gap or a narrow gap for a 3.0 cm ripple? For sound, we will need numbers next lesson." Do NOT calculate the sound or light ratios yet; keep that for Lesson 8.	Claim-Evidence-Reasoning (CER) with Rule Building: learners write the claim from data and justify it with a mechanism. Comparing the prediction with the evidence supports conceptual change, and linking back to $v = f\lambda$ strengthens the connection between wavelength and frequency from earlier lessons.	Listen for statements such as 'the smaller the gap compared with the wavelength, the more the waves spread'. Look for correct use of the term wavelength rather than wave size or wave height. Success criteria: CER includes a data point as evidence, states the rule accurately (comparable to or larger than the gap), and correctly calculates $\lambda = 3.0$ cm from $v = f\lambda$. Misconception to flag: learners saying that waves 'squeeze through' a small gap and so spread, or that bigger waves always spread less.
Driving	Learners return to the class	Say: "Look at the board. Which of	Question Clustering on a Driving	Check that questions are testable

Question Board (DQB) Creation  VIDEO: Reflection and Diffraction of Mechanical Waves Two-Dimensions - Overview Source: CK-12  Search ARES for similar videos  EXERCISE: Wave Interference Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Driving Question Board: 'How can we hear a matatu around a corner that we cannot see, and why does its pitch change and its radio signal fade as it moves?' In groups they write on sticky notes (a) what they now know about the corner puzzle from today's evidence, and (b) new questions that today's evidence raises, for example: 'What are the wavelengths of the matatu music and of light?', 'Does a higher pitch bend less than a lower pitch?', 'Does the 4.0 m opening count as wide or narrow for sound?'. They place their notes under the Part A (corner/diffraction) column and mark any earlier questions that are now partly answered with a tick and a date.	our earlier questions did today's evidence help us answer, even partly?" Give WAIT TIME of 10-15 seconds before asking groups to move their notes. Cold-call 4 groups to read one new question aloud, and ask the class: "Which of these can we test with numbers? What measurement do we need?" Group similar questions together and label clusters (for example, 'sound vs light', 'pitch and bending'). Point out that the questions about sound and light wavelengths will be answered in Lesson 8, and that questions about pitch change and radio fading belong to later lessons. Keep the board visible and add today's date.	Question Board: sorting learner questions into categories makes progress visible and shows how today's evidence connects to the storyline. It gives learners ownership of the investigation and signals what evidence must be gathered next.	and linked to today's rule (for example, mention wavelength, gap or sound versus light) rather than vague (such as 'why is sound different?'). Count how many groups propose a question that requires a numerical comparison of wavelength with the 4.0 m gap. Indicator of understanding: learners tick off at least one earlier question as partly answered and justify it with evidence from the ripple tank or PhET.
Model Building Phase  VIDEO: Properties of periodic waves <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Single Slit Diffraction (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners revisit the class model with its three parts: A (corner/diffraction), B (pitch/Doppler), and C (signal fading/interference). Today they revise only Part A. In groups they draw the street from above: the building, the 4.0 m opening, the parked matatu and Wanjiru and Otieno around the corner. They add curved wavefronts spreading beyond the edge of the wall to show diffraction, and write a rule box beneath the diagram: 'Diffraction is greatest when the wavelength is comparable to or bigger than the gap; waves with a wavelength much smaller than the gap travel almost straight.' They add a note about what is still missing: 'We need the wavelengths of the matatu music and of light to decide which case applies.' They leave Parts B and C	Say: "Take out your model from Lesson 1 and Lesson 2. What did we say about the torch beam and the music? What can we add now?" Ask groups to include at least one labelled wavefront and the 4.0 m gap on the diagram. Circulate and prompt: "Where would the waves be if there were no diffraction? Where are they now?" WAIT TIME of 10-15 seconds before groups respond. Cold-call 3 groups to present their Part A revisions in 30 seconds each, and invite the class to say one thing they agree with and one thing to improve. Remind learners: "We do not yet know the wavelengths for sound and light compared with 4.0 m. That is our job in Lesson 8." Collect the models for display.	Cumulative Model Revision with Gaps Identified: learners update a shared model, include the mechanism (wavefront spreading) and explicitly list what remains unexplained. Naming gaps in the model models scientific honesty and sets up the next lesson's investigation.	Success criteria for Part A: (1) diagram shows curved wavefronts spreading beyond the wall edge; (2) the 4.0 m opening is labelled; (3) the rule on wavelength versus gap is stated correctly; (4) at least one unresolved question is listed. Look for learners who avoid claiming that sound definitely bends and light does not without evidence, since the numbers are still to be compared. Quick exit ticket: 'Complete the sentence: Diffraction is greatest when ____.'

	unchanged except for noting questions that remain.			
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D. TEACHER REFLECTION
Which learners still believe that waves can only travel in straight lines, and what evidence from the ripple tank changed (or did not change) their thinking? Did groups control one variable at a time, and were the ratio calculations accurate? Did learners confuse the size of the gap with the size of the wave or its amplitude? How well did learners avoid jumping to the sound-versus-light conclusion before the numbers were compared? What adjustments to the questioning and wait time would help quieter learners contribute in Lesson 8?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In the ripple tank at 10 Hz ($\lambda = 3.0$ cm), waves passing through a wide gap of 12 cm mostly travelled straight, while waves passing through a narrow gap of 3 cm or 1.5 cm spread out in almost circular wavefronts. The PhET simulation showed the same pattern when slit width or wavelength was changed. The street opening around the building is 4.0 m wide.
What did I learn?	Diffraction is the spreading of waves around an obstacle or through a gap. It is greatest when the wavelength is comparable to or bigger than the gap, and is very small when the wavelength is much smaller than the gap. The wavelength can be found from $v = f\lambda$, for example $0.30 \div 10 = 0.03$ m = 3.0 cm.
How does this explain the phenomenon?	This partly explains Part A of the matatu puzzle: waves can bend around the corner if their wavelength is comparable to the 4.0 m opening. We still need to compare the wavelengths of sound and light with 4.0 m in Lesson 8 to explain why the music is heard around the corner but the torch beam is blocked.

LESSON 8 (40 minutes (one lesson); can be extended to a double lesson of 80 minutes for the full model revision): Diffraction and the Corner Puzzle: Why Sound Turns the Corner but Light Does Not

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To use the diffraction rule from Lesson 7 to compare the wavelengths of sound and light with the 4.0 m street opening, so learners can explain why Wanjiru and Otieno hear the matatu music around the building while a torch beam from the matatu is blocked by the wall.
Knowledge	Learners will be able to: (a) calculate the wavelength of sound from the wave equation $v = f\lambda$ using $v = 340$ m/s (400 Hz gives 0.85 m, 200 Hz gives 1.7 m, 1000 Hz gives 0.34 m); (b) compare each wavelength with the 4.0 m opening by calculating the ratio λ/gap ($0.85/4.0 = 0.21$ for the 400 Hz horn); (c) explain that sound with wavelength 0.34 m to 1.7 m is comparable to the opening and diffracts clearly, while visible light of 5.0×10^{-7} m (ratio about 1.3×10^{-7}) hardly diffracts; (d) state that low pitches (long wavelength) bend around the corner more than high pitches (short wavelength).
Skills	NGSS Science and Engineering Practices: Using Mathematics and Computational Thinking (calculating wavelengths and ratios with $v = f\lambda$); Analysing and Interpreting Data (comparing ratios in a table); Constructing Explanations (using the diffraction rule as the reasoning); Developing and Using Models (revising Part A of the class model). Core competencies: Critical thinking and problem solving; Communication and collaboration; Digital literacy (PhET Sound Simulation).
Attitudes	Values: Responsibility (each group member plays an assigned role: calculator, recorder, reporter, checker) and Respect (listening to peers' reasoning and disagreeing with evidence, not opinion). Curiosity about everyday sound and light in Nairobi streets. PCI: Safety and Security Education (stay on the pavement and never stand in the road when observing matatus; keep sound volumes safe for ears).
Key Inquiry Question	1. How do waves behave when they meet obstacles or different media? 2. How are the properties of waves applied in day-to-day life? (Focus today: how do sound and light waves behave when they meet the wall and the 4.0 m opening, and why do they behave differently?)
Purpose in Storyline	Lessons 1 and 2 raised the question of why the torch beam is blocked by the wall but the music is heard; Lesson 7 gave the rule that diffraction is greatest when wavelength is comparable to or larger than the gap. Lesson 8 applies that rule with real numbers to the 4.0 m opening and completes the evidence for Part A (corner/diffraction) of the class model. It leaves Part B (pitch change, taught in Lesson 11) and Part C (signal fading, taught in Lesson 9) open and added to the DQB.
Safety Notes	No hazardous materials. Keep speaker or phone volume moderate (never above a comfortable conversation level) during the sound demonstration. Do not look directly into the torch beam, and do not shine it into anyone's eyes. If doing the corner walk outdoors, stay within the school compound, away from the road, and keep learners in sight of the teacher.

B. LESSON OVERVIEW

Mr Kamau returns to the puzzle that Wanjiru and Otieno met on the Nairobi street: they cannot see the matatu behind the building, yet they hear its music clearly, while a torch beam from the matatu would be blocked by the wall. Learners first predict which wavelengths will bend around the corner and whether high or low pitches will be heard better. In the Observe Phase they gather evidence in three ways: a short corner demonstration with a speaker playing a low tone and a high tone behind a classroom partition (or door) beside a torch; a calculation table where they use the wave equation $v = f\lambda$ with 340 m/s to find the wavelengths of the horn (400 Hz, 0.85 m) and the music (200 Hz, 1.7 m and 1000 Hz, 0.34 m), then compare each with the 4.0 m opening, giving ratios such as $0.85/4.0 = 0.21$; and a comparison with visible light of wavelength 5.0×10^{-7} m, whose ratio to the opening is about 1.3×10^{-7} , essentially no bending. In the Explain Phase learners write a claim-evidence-reasoning explanation of why sound turns the corner and light does not, and explain why high pitches bend less than low pitches. They add a question to the Driving Question

Board and revise Part A of the class model. Parts B (pitch/Doppler) and C (signal fading/interference) are only flagged as still open.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Equation (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the street scene: Wanjiru and Otieno stand around the corner, the matatu is hidden behind the concrete wall, and the 4.0 m street opening is the gap the sound passes through. Individually, in their notebooks, they answer three prediction prompts: (1) Will the 1000 Hz music bend around the corner as well as the 200 Hz music? (2) Which will bend more, the sound or a torch beam? (3) What do you think decides how much a wave bends? They sketch a top-view diagram of the building, the 4.0 m opening, the matatu and the students, drawing arrows for how they think sound and light travel. They then share with a partner and note any disagreements for later testing.	Display a top-view diagram of the street on the board. Say: "Wanjiru and Otieno can hear the matatu but cannot see it. In Lesson 7 we found a rule about wave size and gap size. Before I give you any numbers, predict: will a high pitch or a low pitch be heard better around the corner? Write your prediction and one reason." Give 2 minutes of silent writing, then WAIT TIME of 10 to 15 seconds after the question before taking any answer. Pair-share for 1 minute. Cold-call 4 learners (2 who predict low pitch, 2 who predict high pitch) and record their predictions in two columns on the board without judging. Say: "Hold on to these predictions, we will test them with evidence today." Circulate and note any learners who say that sound simply 'goes everywhere' so you can probe their reasoning in the Explain Phase.	Prediction with reasoning (Predict-Observe-Explain). Writing a prediction and a reason before seeing data surfaces prior ideas, such as 'sound is stronger than light' or 'sound goes everywhere', and makes learners accountable to evidence. Returning to the predictions later lets learners see how their thinking changed, which deepens conceptual change.	Look for: (a) learners who connect their prediction to wavelength or the size of the gap, which shows they are using the Lesson 7 rule; (b) learners who give vague reasons like 'sound is louder', which flags a misconception to address later; (c) sketches that show sound arrows curving or spreading past the corner versus a straight torch arrow. Tally predictions on the board (low vs high pitch) to use as a baseline for comparison after the Observe Phase.
Observe Phase  VIDEO: Sound Properties: Amplitude, period, frequency, wavelength Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners work in groups of four with assigned roles (calculator, recorder, reporter, checker) in three stations of about 5 minutes each. Station 1, Corner demonstration: Mr Kamau stands behind a classroom door opened about 1 m (or a partition) with a phone or speaker playing a low tone (about 200 Hz) then a high tone (about 1000 Hz) at equal comfortable volume. Learners sit at an angle in the room where the speaker is not in direct line, and	Before the stations say: "Your job is to collect evidence to test your predictions. Every number must go in the table." At Station 1, ask: "Where were you sitting relative to the speaker? Which tone could you hear more clearly? How did the torch behave in the same spot?" At Station 2, model one calculation on the board: "$v = f\lambda$ so $\lambda = v$ divided by f; for the horn, 340 divided by 400 equals 0.85 m." Ask: "What do you notice about the wavelength when the	Data table and pattern-finding (Analysing and Interpreting Data). Putting frequency, wavelength, ratio and amount of bending side by side lets learners spot that smaller ratios (λ/gap) mean more bending, and that the light ratio is millions of times smaller than the sound ratios. Combining a physical demonstration with calculation links the observation (hearing around the corner) to the quantitative rule.	Check that learners: (a) correctly compute 0.85 m, 1.7 m and 0.34 m using $v = f\lambda$ with 340 m/s; (b) correctly find the ratio $0.85/4.0 = 0.21$; (c) record that the low tone is heard more clearly than the high tone out of line of sight; (d) write the light wavelength as 5.0×10^{-7} m and recognise that its ratio is far smaller than for sound. Use a quick checklist for each group. Ask the checker in each group to initial the table to confirm calculations before moving on.

 EXERCISE: Wave Equation Practice Source: CK-12  Search ARES for similar readings	record which tone sounds clearer, and then try the torch from the same position and note that the beam does not reach them. Station 2, Wavelength table: using $v = f\lambda$ with $v = 340$ m/s they calculate λ for 200 Hz (1.7 m), 400 Hz horn (0.85 m) and 1000 Hz (0.34 m), then divide each by the 4.0 m opening, getting 0.43, 0.21 and 0.085. Station 3, Light comparison: they take light of wavelength 5.0×10^{-7} m, compare it with 4.0 m, compute the ratio of about 1.3×10^{-7}, and compare it with the sound ratios. They may also use the PhET Sound Simulation to watch waves spread from a source of different frequencies. They complete a class data table of source, frequency, wavelength, ratio to opening, and amount of bending (large, medium, almost none).	frequency goes up?" WAIT TIME of 10 to 15 seconds before taking answers. At Station 3, ask: "How many times smaller is the light wavelength than the sound wavelength? What does a ratio of about 1.3×10^{-7} tell us about bending?" Check units (metres, hertz, m/s) and that learners write 5.0×10^{-7} m correctly. Cold-call 3 groups to report their ratio for the 400 Hz horn (0.21) and verify the class agrees. Remind learners: "Stay seated and keep the torch away from faces. Keep the speaker volume low." Where the corner demonstration is hard to hear, move the speaker closer or repeat it.		
Explain Phase  VIDEO: Reflection and Diffraction of Mechanical Waves Two-Dimensions - Overview Source: CK-12  Search ARES for similar videos  HTML: Wave Interference Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners use their data table to write a Claim-Evidence-Reasoning (CER) explanation in their groups: Claim: why Wanjiru and Otieno hear the matatu music around the building but a torch beam would be blocked. Evidence: the wavelengths 0.34 m to 1.7 m compared with the 4.0 m opening (ratios 0.085 to 0.43, for the horn 0.21), and light at 5.0×10^{-7} m (ratio about 1.3×10^{-7}); and the classroom corner observation. Reasoning: the diffraction rule from Lesson 7, that diffraction is greatest when wavelength is comparable to or bigger than the gap. They then extend the explanation to answer: why do the low pitches of the music bend more than high pitches? (a longer wavelength is closer to the size of the opening). They compare their	Say: "Use the three-part frame: claim, evidence, reasoning. Your evidence must include at least two numbers from the table." Display sentence starters: "The sound bends because the wavelength of ___ m is ___ compared with the 4.0 m opening..." and "Light does not bend noticeably because...". Circulate and ask: "Why does the ratio matter, rather than the wavelength alone?" and "If the driver played a 1000 Hz tone only, would the students still hear it around the corner? How well?" WAIT TIME of 10 to 15 seconds after each question. Then run a gallery swap: groups exchange papers and give one 'glow' and one 'grow'. Cold-call 3 groups to read their reasoning aloud and ask the class to give a thumbs up or down if the reasoning uses the	Claim-Evidence-Reasoning (Constructing Explanations) with peer critique. Structuring the explanation forces learners to link numerical evidence to a scientific principle rather than just stating a fact. Peer review exposes gaps such as missing numbers or reasoning that mentions only wavelength without the gap size.	Use a 3-point rubric: (1) the claim answers the corner puzzle for sound and light; (2) the evidence cites at least two correct numbers (for example 0.85 m, 4.0 m, 0.21 and 5.0×10^{-7} m); (3) the reasoning states that diffraction is greatest when wavelength is comparable to or bigger than the gap. Look for learners who explain why low pitch bends more than high pitch. Note any who confuse wavelength with frequency or say that light 'cannot' bend at all, and follow up with the group.

	written claims with another group and revise. Finally, they revisit the Predict Phase tallies and mark which predictions were supported.	Lesson 7 rule. Address the misconception that light does not diffract at all: say, "Light does diffract, but its wavelength of 5.0×10^{-7} m is so tiny compared with 4.0 m that the bending is far too small to notice. That is why the torch beam looks blocked." Do not teach the Doppler effect or interference here; if learners ask about pitch change or fading, say, "Great question. Let's put it on the board; we will test it in later lessons."		
Driving Question Board (DQB) Creation VIDEO: Wave properties Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Wave Equation (Flexbook) Source: CK-12 Search ARES for similar readings	Learners return to the Driving Question Board with the question: How can we hear a matatu around a corner that we cannot see, and why does its pitch change and its radio signal fade as it moves? In groups they write on sticky notes (a) what they now know about the corner puzzle, and (b) new or remaining questions. They place answered questions in the 'Answered' column with evidence (for example, 'Sound of 0.85 m is comparable to the 4.0 m opening, so it diffracts: ratio 0.21'), and place unanswered questions in 'Still to investigate' (for example, 'Why is the horn higher pitched while approaching?', 'Why does the FM radio fade near the Ngong Hills?'). They may add new questions such as 'Do radio waves of 3.0 m bend around hills?'.	Say: "Look at the board. Which of our questions have we answered with evidence today? Which ones are still open?" Give each group 3 minutes to write notes. WAIT TIME of 10 to 15 seconds before the first group shares. Cold-call 3 groups to move one sticky note to the 'Answered' column and justify it using a number from today's table. Ask: "What evidence would we need to answer the pitch question?" and "What do the weak radio spots tell us about the signal?" Do not answer these; just record them as questions for the next lessons. Affirm good questions by saying, "That is a testable question."	Driving Question Board as a public record of growing understanding. Sorting questions into answered and open shows learners that science progresses through evidence, builds ownership of the storyline, and connects today's finding (Part A) to the next lessons on interference and the Doppler effect.	Look for sticky notes that cite specific evidence (numbers, ratios) rather than general statements. Check that learners correctly sort the corner question as answered and the pitch and fading questions as open. Note whether new questions are testable and linked to the phenomenon, and whether any learners are still confusing the three puzzles.
Model Building Phase VIDEO: Sound Properties: Amplitude, period, frequency.	Learners revise Part A (corner/diffraction) of the one class model. Individually, then in groups, they update their model diagram: a top-view drawing of the building, the 4.0 m opening, the matatu and the two students, showing wavefronts of sound with wavelength 0.85 m spreading and	Say: "Revise only Part A today. Your model must show why sound bends and light does not, using the numbers." Give 8 minutes of work time. Circulate and ask: "Where in your drawing can I see the wavelength compared with the opening?" and "What would change in your model if the matatu	Cumulative model revision (Developing and Using Models). Updating a visual model with quantitative evidence shows how explanations grow stronger over time. Leaving Parts B and C open shows what the model cannot yet explain, which motivates the next investigations.	Check that each model: (a) shows curved sound wavefronts around the corner and a blocked straight light ray; (b) uses correct values 0.85 m, 0.34 m to 1.7 m, 4.0 m, 0.21 and 5.0×10^{-7} m; (c) states that low pitch bends more than high pitch; (d) leaves Parts B and C unfinished. Use the exit ticket to

wavelength Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Wave Equation Practice Source: CK-12 Search ARES for similar readings	curving around the corner, and a straight arrow for light of wavelength 5.0×10^{-7} m blocked by the wall. They add labels for the numbers: $\lambda = 0.85$ m for the 400 Hz horn, λ from 0.34 m to 1.7 m for the music, ratio 0.21, light ratio about 1.3×10^{-7}, and a note that lower pitches bend more than higher pitches. They write a two-sentence caption explaining Part A in their own words. For Parts B and C they leave labelled blank boxes titled 'Pitch change (to be explained)' and 'Signal fading (to be explained)'.	played only a 1000 Hz tone?" WAIT TIME of 10 to 15 seconds before learners respond. Cold-call 3 groups to share their revised Part A and ask others to identify one improvement from their earlier model. Build the class model on a chart by combining the best features: a labelled drawing and a numerical caption. Remind learners to keep Parts B and C blank for later lessons. Finish with an exit ticket: "In one sentence, explain why light from the matatu is blocked by the wall but its music is heard around the corner, using the wavelength and the 4.0 m opening."	sort responses into: complete (uses wavelength and gap size), partial (mentions wavelength only), or needs support (says sound is stronger than light). Plan a short revisit for the third group.
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D. TEACHER REFLECTION

After the lesson, consider: Did learners use the ratio of wavelength to gap (not wavelength alone) in their reasoning? Which learners still believe that light cannot diffract, and what evidence helped or did not help them? Was the corner demonstration audible enough for all learners, and should the tone volume or seating be adjusted? How well did the groups manage the calculations with $v = f\lambda$ and units? Did any learners start explaining the pitch change or the fading radio signal too early, and how did you redirect them? Which questions on the Driving Question Board show the strongest curiosity that you can use to launch Lesson 9 on interference and the fading radio? Note any learners needing extra support with scientific notation (5.0×10^{-7} m) before the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a low tone (about 200 Hz) was heard more clearly than a high tone (about 1000 Hz) around the corner of the partition, while the torch beam did not reach us. We calculated that the matatu horn (400 Hz) has wavelength 0.85 m, the music ranges from 1.7 m (200 Hz) to 0.34 m (1000 Hz), and light has wavelength 5.0×10^{-7} m, compared with the 4.0 m street opening.
What did I learn?	We learned that diffraction is greatest when the wavelength is comparable to or bigger than the gap. For the horn, the ratio of wavelength to gap is $0.85/4.0 = 0.21$, so the sound clearly bends, while for light the ratio is about 1.3×10^{-7} , so it hardly bends at all. Lower pitches (longer wavelengths) bend around the corner more than higher pitches.
How does this explain the phenomenon?	Wanjiru and Otieno hear the matatu music around the building because the sound waves of wavelength 0.34 m to 1.7 m are comparable to the 4.0 m opening and diffract around the corner, while the torch beam, with wavelength 5.0×10^{-7} m, hardly diffracts and is blocked by the wall. The pitch change and the fading radio signal are still open questions on our Driving Question Board.

LESSON 9 (40 minutes (double lesson of 80 minutes recommended if the speaker walk is done outdoors)): Interference and the Fading Radio: Loud Spots, Quiet Spots and Weak FM Signal

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how two overlapping waves combine, so they can explain the loud and quiet spots between two speakers and the fading FM signal near the Ngong Hills (Part C of the class model).
Knowledge	Superposition is the adding together of overlapping waves. Interference is the result. Constructive interference (reinforcement) occurs when path difference = $n\lambda$. Destructive interference (cancellation) occurs when path difference = $(n+1/2)\lambda$. Two speakers playing 340 Hz have $\lambda = 340/340 = 1.00$ m, so a path difference of 1.00 m gives a loud spot and 0.50 m gives a quiet spot. FM at 100 MHz has $\lambda = 3.0 \times 10^8 / 1.0 \times 10^8 = 3.0$ m. Direct and hillside-reflected signals cancel at weak spots half a wavelength apart, 1.5 m.
Skills	Predicting, observing and recording patterns in a ripple tank; planning and carrying out a walk across a sound field; calculating λ using the wave equation $v = f\lambda$ and path difference; using evidence to construct an explanation (NGSS SEP: Planning and Carrying Out Investigations; Analyzing and Interpreting Data; Constructing Explanations).
Attitudes	Curiosity about everyday signal problems, collaboration in teams, respect for different ideas when predictions are tested, and persistence when results differ from expectations.
Key Inquiry Question	1. How do waves behave when they meet obstacles or different media? 2. How are the properties of waves applied in day-to-day life?
Purpose in Storyline	Lessons 7 and 8 explained why the music turns the corner (diffraction). Lesson 9 answers the third puzzle: why the FM signal fades at certain spots near the Ngong Hills even though the station never changes. It supplies the evidence for Part C of the class model and prepares Lesson 10 (standing waves), which links the 1.5 m spacing to half a wavelength.
Safety Notes	Keep speaker volume moderate, since 340 Hz tones can be tiring and loud tones can harm hearing. Wipe up spilled water around the ripple tank at once to prevent slipping and keep mains-powered equipment dry. Keep learners walking slowly with clear paths when they walk across the room with their eyes on the floor. Switch off the signal generator after use.

B. LESSON OVERVIEW

Mr Kamau begins by reminding the class of Wanjiru and Otieno's three puzzles. Learners have explained the corner (Lesson 8), so today they tackle the fading FM radio in Baraka's matatu on the Nairobi-Ngong route. Learners first predict what happens when two sets of waves overlap. They then observe two ripple sources in a ripple tank, then walk in front of two speakers both playing 340 Hz and map loud and quiet spots. In the Explain phase they use $v = f\lambda$ to find $\lambda = 1.00$ m, link loud and quiet spots to path differences of 1.00 m and 0.50 m, and then scale the idea to the 100 MHz FM station ($\lambda = 3.0$ m) whose direct and hillside-reflected signals cancel at weak spots 1.5 m apart. Learners add questions to the DQB and build Part C of the class model. The lesson does not teach standing waves (Lesson 10) or the Doppler effect (Lesson 11).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Constructive and Destructive interference Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Interference Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners recall the third puzzle: Baraka's FM radio fades and crackles at certain spots near the Ngong Hills and comes in clearly a few metres on, though the station never changes. Individually, learners draw two sets of circular ripples spreading from two dippers and predict in their notebooks what happens where they overlap. Then they predict: 'If two speakers both play the same 340 Hz tone, will the sound be equally loud everywhere in front of them?' They sketch a top-view of the room with predicted loud (L) and quiet (Q) spots. Pairs compare predictions, then share with the class.	Say: 'Remember Baraka's radio. The station is always 100 MHz. So why would the signal be strong here and weak one step later?' Hold up two stones or draw two dippers on the board. Ask: 'What do you think happens when two waves meet?' Give 10-15 seconds of WAIT TIME before anyone speaks, then cold-call 4 learners to share predictions without judging. Record ideas such as 'they add', 'they cancel', 'they bounce off each other' on the board under 'Our predictions'. Ask: 'What would make you change your mind?'	Predict-Observe-Explain: committing to a written prediction first creates a cognitive conflict that makes learners care about the evidence, and the recorded misconceptions (for example, 'waves bounce off each other') can be revisited and corrected after observation.	Check notebooks for: (1) a drawn overlapping pattern; (2) a stated prediction about loudness (equal everywhere or not); (3) a reason given. Tally on the board how many learners predict equal loudness everywhere versus varying loudness. Note who holds the idea that waves collide and stop.
Observe Phase  VIDEO: Constructive and Destructive interference Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Interference (Flexbook) Source: CK-12  Search ARES for similar readings	Station 1 (ripple tank, 10 minutes): Two dippers on the vibrator produce ripples at 10 Hz. Learners look at the pattern and describe lines of strong disturbance and lines of calm water spreading out from between the sources. They sketch the pattern and mark where crests meet crests and where crests meet troughs. Station 2 (speakers, 12 minutes): Two speakers, side by side, both play 340 Hz from one signal generator. Learners in groups of four take turns: one walks slowly in a straight line about 3 m in front of the speakers with one ear covered, while the others mark with chalk or paper the spots where the sound is loud (L) or quiet (Q). They measure the distance between neighbouring loud spots and between a loud and a quiet spot with a tape measure and record it in a table. They also cover one speaker and note that the quiet spots disappear. Optional: the PhET Sound	Circulate and ask: 'Where is the water almost flat? What are the two ripples doing there?' and 'Point to a quiet spot. Is the sound dead, or is a speaker still working?' Prompt: 'Close one ear and walk slowly; mark the spot as soon as the loudness changes.' Ask groups to cover one speaker and listen again, and ask: 'What happened to the quiet spots? What does that tell you?' Use WAIT TIME of 10-15 seconds after each question. Cold-call 3 groups to report the spacing they measured. Remind learners to keep the volume moderate. Say: 'Remember that the signal generator frequency is 340 Hz. Keep that number, we need it in the Explain phase.'	Structured data collection with pattern spotting: learners gather the same observations in two different media (water and sound) so they notice that loud and quiet spots are a property of waves in general, not of one machine.	Look for: (1) sketches that show alternating bands of strong and calm water spreading from between the sources; (2) a table with at least 4 marked spots and measured spacing; (3) learners noticing that covering one speaker removes the quiet spots. Probe learners who say 'the speaker is broken at the quiet spot'.

	simulation or LabXchange Wave Interference can be used if equipment is limited.			
Explain Phase  VIDEO: Beat frequency <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Wave Interference Practice Source: CK-12  Search ARES for similar readings	Learners use their data in four steps. Step 1: use the wave equation $v = f\lambda$ with $v = 340$ m/s and $f = 340$ Hz to get $\lambda = 340/340 = 1.00$ m. Step 2: explain the speakers. For a spot equidistant from both speakers, the path difference is 0, waves arrive crest with crest, and they reinforce (constructive). They are told that at a spot where the path difference is 1.00 m (one whole λ) the waves still arrive in step and the spot is loud, while at a spot where the path difference is 0.50 m (half λ) a crest arrives with a trough, the waves cancel and the spot is quiet (destructive). They write the rules: constructive when path difference = $n\lambda$; destructive when path difference = $(n+1/2)\lambda$. Step 3: they complete a table for path differences of 0, 0.50 m, 1.00 m, 1.50 m and 2.00 m and label each loud or quiet. Step 4: transfer to the FM radio. They calculate λ for the 100 MHz station: $\lambda = 3.0 \times 10^8 / 1.0 \times 10^8 = 3.0$ m. Near the Ngong Hills, the signal that comes directly to the matatu and the signal reflected from the hillside overlap. As the matatu moves, the extra distance travelled by the reflected signal changes, so the path difference changes between $n\lambda$ and $(n+1/2)\lambda$. The weak spots, where the path difference changes by half a wavelength, are 1.5 m apart. Each learner writes a 3-4 sentence explanation of why Baraka's radio fades and returns while the station never changes, and draws a labelled diagram of the direct and	Say: 'Use the wave equation. What are v and f for the speakers?' Cold-call 2 learners for λ and wait 10-15 seconds. Draw two wave paths on the board, one with a 1.00 m difference and one with 0.50 m. Ask: 'If the path differs by exactly one wavelength, do the crests line up or not?' then cold-call 3 pairs. Say: 'Why is the quiet spot not silent everywhere? Think about the two speakers.' Introduce the rules $n\lambda$ and $(n+1/2)\lambda$ only after learners have described the pattern in their own words. Challenge: 'Radio waves are not sound. Do you think the same rule applies? What would the wavelength be?' Guide the calculation of $\lambda = 3.0$ m and show 1.5 m as half of it. Say clearly that this lesson does not explain why the pattern forms a fixed standing pattern, which is for Lesson 10. Warn against the misconception that 'the signal got weaker because of distance': 'The station power did not change. What changed is how the two signals add up.'	Claim-Evidence-Reasoning with analogy transfer: learners make a claim about the fading radio, support it with speaker evidence and calculations, and reason with path difference. Transferring the same idea from sound ($\lambda = 1.00$ m) to radio ($\lambda = 3.0$ m) strengthens understanding that interference is a general wave property.	Success indicators: (1) $\lambda = 1.00$ m calculated correctly with units; (2) learners match path difference 1.00 m to a loud spot and 0.50 m to a quiet spot; (3) the FM wavelength of 3.0 m and the 1.5 m spacing are correct; (4) the explanation mentions two signals (direct and reflected) overlapping and cancelling, not 'the station got weak'. Use a 3-question exit check on mini whiteboards: 'Path difference 1.00 m, loud or quiet?' 'Path difference 0.50 m?' 'What is the spacing of FM weak spots?'

	reflected paths.			
Driving Question Board (DQB) Creation  VIDEO: Properties of periodic waves <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Interference Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners revisit the Driving Question Board and use sticky notes to (a) place a green dot on questions now answered (for example, 'Why does the radio signal fade at certain spots?'), (b) add new questions that arose (for example, 'Why do the fading spots stay in the same places?', 'Do the weak spots move if the matatu stops?', 'Could we get a quiet spot with a different frequency?', 'Why does the pitch change?'), and (c) group new questions under headings: still about interference, about standing patterns, about pitch change.	Say: 'Look at the DQB. Which questions can we now answer with evidence from today?' Give 10-15 seconds WAIT TIME, then cold-call 3 learners to move a question and justify it using speaker or ripple data. Invite 3 more to read out new questions. Say: 'Which of these questions will we explore in the next lessons?' Mark the fixed-pattern question as the bridge to Lesson 10 and the pitch question for Lesson 11.	Question tracking on the DQB: showing which questions are answered by evidence and which remain makes progress visible and keeps learners connected to the anchoring phenomenon.	Observe whether new questions use today's vocabulary (path difference, cancel, reinforce) and whether learners cite evidence for moving a question. Note any question showing a misconception, for example 'does the radio break at the weak spot?', for follow-up.
Model Building Phase  VIDEO: Properties of periodic waves - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Wave Interference (Flexbook) Source: CK-12  Search ARES for similar readings	Learners revise the class model, adding Part C (signal fading/interference) while Parts A (corner/diffraction) and B (pitch/Doppler) stay on the display. Part C is drawn as a diagram of the matatu on the Nairobi-Ngong road, a direct signal arrow from the transmitter, a reflected signal arrow from the Ngong Hills, and labels: '$\lambda = 3.0 \text{ m}$', 'constructive: path difference = $n\lambda$, strong signal', 'destructive: path difference = $(n+1/2)\lambda$, weak signal', 'weak spots 1.5 m apart'. Part B is left with a note: 'Not yet explained'. In groups they add a small inset of the two-speaker loud/quiet pattern as supporting evidence and write one sentence connecting the evidence to the model. They then complete	Say: 'Our model needs a Part C. What must be in it to explain Baraka's radio?' Give 10-15 seconds WAIT TIME, then cold-call 4 learners, each adding one element. Ask: 'What evidence do we have for each label?' Point to the speaker data. Ask: 'Does the model say the radio station changed?' (No.) Check that groups use the 3.0 m and 1.5 m figures exactly. Say: 'Next lesson we will see what happens when waves travel in opposite directions along a guitar string and meet, and why the 1.5 m spacing is half a wavelength.'	Cumulative model building: each lesson adds one part of a single class model, so learners see how separate properties of waves (diffraction, interference) combine to explain one real event.	Use a model checklist: (1) direct and reflected paths drawn; (2) $\lambda = 3.0 \text{ m}$ shown; (3) both interference rules stated correctly; (4) 1.5 m spacing labelled; (5) statement that the station does not change. Groups with 4 or 5 ticks are secure; those with fewer need a short conference before Lesson 10.

	the summary table.			
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D. TEACHER REFLECTION

Which learners still believed that waves 'bounce off' or 'destroy' each other, and did the covered-speaker test change their minds? Were learners able to calculate $\lambda = 1.00\text{ m}$ and $\lambda = 3.0\text{ m}$ unaided? Did the walk across the sound field give clear loud and quiet spots despite echoes from classroom walls, and how could the setup be improved (for example, moving the activity outside or to the school field)? Which learners confused interference with a weakening of the source, and what additional evidence might help? Did the DQB questions suggest learners are ready for Lesson 10?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In the ripple tank, two sources made bands of strong ripples and calm water. With two speakers playing 340 Hz we found loud and quiet spots in the room, and the quiet spots vanished when one speaker was covered.
What did I learn?	Overlapping waves add together (superposition). When the path difference is $n\lambda$ the waves reinforce (constructive interference, loud or strong). When it is $(n+1/2)\lambda$ they cancel (destructive interference, quiet or weak). For 340 Hz, $\lambda = 1.00\text{ m}$, so a path difference of 1.00 m gives a loud spot and 0.50 m gives a quiet spot.
How does this explain the phenomenon?	Near the Ngong Hills, the 100 MHz FM signal has $\lambda = 3.0\text{ m}$. The direct signal and the signal reflected from the hillside overlap and cancel at weak spots about 1.5 m apart, then reinforce between them. The radio fades and returns even though the station never changes.

LESSON 10 (40 minutes (double lesson of 80 minutes recommended if the model-building phase is extended)): Standing Waves and Musical Notes: Guitar Strings, Nodes and the Half-Wavelength Link

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how two waves travelling in opposite directions along a string superpose to form a standing wave, and use this to explain how string length and tension set the musical note. They then link the half-wavelength spacing in a standing wave to the 1.5 m spacing of the FM fading spots near the Ngong Hills.
Knowledge	Learners will be able to: (a) explain that a standing wave forms when a wave and its reflection from a fixed end travel in opposite directions and superpose; (b) identify nodes (no movement) and antinodes (maximum movement) and state that adjacent nodes are $\lambda/2$ apart; (c) use $L = \lambda/2$ for the fundamental and $v = f\lambda$ to show that the 0.65 m guitar string with wave speed 143 m/s has fundamental wavelength 1.30 m and frequency 110 Hz; (d) show that the second harmonic has wavelength 0.65 m and frequency 220 Hz (3 nodes, 2 antinodes), and that pressing the string to 0.325 m at the same tension also gives 220 Hz; (e) link the 1.5 m spacing of the weak FM spots to half the 3.0 m wavelength.
Skills	Developing and using models (sketching standing wave patterns); planning and carrying out investigations (varying length and tension one at a time); analysing and interpreting data (comparing predicted and measured frequencies); using mathematics ($v = f\lambda$, $L = \lambda/2$, $T = 1/f$); constructing explanations; communication and collaboration.
Attitudes	Curiosity about how musical instruments such as the nyatiti, guitar and litungu produce notes; responsibility when handling tensioned strings; respect for the ideas of others during group discussion; appreciation of physics in music and communication in Kenya.
Key Inquiry Question	How do waves behave when they meet obstacles or different media? How are the properties of waves applied in day-to-day life?
Purpose in Storyline	Lessons 8 and 9 explained the corner puzzle through diffraction and the fading radio through interference. In Lesson 10 learners see what happens when reflected waves keep overlapping with the original wave: a stable pattern of fixed quiet points (nodes) and strong points (antinodes). This deepens Part C of the class model by showing why the quiet spots are half a wavelength apart, and it adds a real application of waves (musical notes) that answers the second KICD key inquiry question. It also prepares learners for Lesson 11 (the Doppler effect) by reinforcing $v = f\lambda$.
Safety Notes	Keep the slinky on the floor or a smooth table and never release a stretched end suddenly; it can snap back. Keep faces away from the string when tightening a guitar string and do not over-tighten; wear no loose items near the tuning pegs. Keep slinky stretch to about 3 m and secure one end held by a learner or tied to a table leg. Keep walking space clear of cables if using laptops or tablets for the simulation. Keep the volume low when plucking or using the buzzer.

B. LESSON OVERVIEW

Mr Kamau reminds the class of Wanjiru and Otieno standing around the corner, hearing the matatu music and then meeting a fading FM radio near the Ngong Hills. In Lesson 9 learners found that weak spots appear where the direct and reflected radio signals cancel. Today the question is: what happens when a wave and its reflection travel along the same line for a long time? Learners first predict what a guitar string of length 0.65 m will do when plucked. They observe standing waves on a slinky and a guitar string, then use the PhET Standing Waves on a String simulation (offline copy) to count nodes and antinodes and vary frequency, tension and length. In the Explain phase they use $L = \lambda/2$ and $v = f\lambda$ to show that the 0.65 m string with wave speed 143 m/s gives 110 Hz, the second harmonic gives 220 Hz, and a string pressed to 0.325 m also gives 220 Hz. They then connect nodes spaced $\lambda/2$ apart to the weak FM spots 1.5 m apart (half of 3.0 m). The class updates the Driving Question Board and extends Model Part C with standing-wave ideas.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Standing waves on strings Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave on a String Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners sit in groups of four. Mr Kamau plucks the guitar string (length 0.65 m) and then presses it halfway along (0.325 m) and plucks again. Learners record individual predictions in their notebooks: (1) Why does the string seem to vibrate in a fixed pattern with some points hardly moving? (2) Will the shorter string give a higher or lower note? (3) If the string is tightened, what happens to the note? Each learner draws a quick sketch of what the string looks like while vibrating. They then link back to the phenomenon by writing one sentence on how this could relate to Wanjiru and Otieno's fading radio spots, where some positions are weak and others are strong.	Say: "Last lesson we found that two overlapping waves can make quiet spots and loud spots. Today we ask what happens when a wave bounces back along the same line and keeps overlapping with itself." Pluck the long string, then the pressed string. Say: "Before I tell you anything, write your prediction and draw the string." Give silent WAIT TIME of 10-15 seconds after each question. Then cold-call 4 learners from different groups: "What does your sketch show and why?" Ask 2 learners to state their reasoning for shorter versus longer string. Do not correct predictions; write 3 different ideas on the board under the heading 'Our Predictions'. Say: "Keep these. We will test them with evidence."	Predict-and-sketch with public recording of ideas. Learners commit to a model before instruction, which makes later evidence more meaningful because they can compare it with their own thinking and see changes in their reasoning.	Observe whether at least 80% of learners wrote a prediction for both length and tension. Look for the misconception that the whole string moves equally or that only the plucked point moves. Note learners who link nodes to quiet spots, or who say a shorter string gives a higher note. Record the number of sketches that show points of no movement.
Observe Phase  VIDEO: Standing waves on strings Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Standing Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Learners rotate through three stations in groups of four, 5 minutes each. Station 1 (slinky): two learners stretch the slinky to about 3 m on the floor and a third shakes one end sideways at a steady rate while the fourth watches. They find the rate that makes the slinky vibrate in one big loop (1 antinode, 2 nodes at the ends), then double the rate for 2 loops (3 nodes, 2 antinodes). They count nodes and antinodes and draw each pattern. Station 2 (guitar string): the 0.65 m string is plucked at full length, lightly touched at the midpoint (0.325 m) and plucked again; learners listen and compare notes. They then	Say: "You are not memorising today; you are collecting evidence. Count nodes and antinodes carefully." Circulate and ask: "Where does the slinky not move at all? What do we call those points?" At the guitar station ask: "What changed when you shortened the string? What stayed the same?" Give WAIT TIME of 10-15 seconds before accepting answers. At the simulation say: "Change one thing at a time. What did you change, and what did you observe?" Cold-call 3 learners at stations to describe a pattern they saw. Remind learners about safety: "Hold the slinky end firmly, never release it suddenly." Ensure	Evidence gathering with a structured observation table (data-first approach). Learners describe patterns in everyday language before receiving vocabulary, building a meaningful link between what they saw and the scientific terms. Multiple representations (slinky, string, simulation) strengthen the pattern across different media.	Check that learners can: (a) point to nodes and antinodes on their slinky pattern; (b) count 2 nodes and 1 antinode for the first pattern and 3 nodes and 2 antinodes for the second; (c) state that the shorter string gave a higher note; (d) state that tighter string gave a higher note. Use a tally on the board of groups with correct tables. Listen for learners confusing nodes with antinodes and re-ask guiding questions.

	tighten the string slightly and note the pitch change. Station 3 (PhET Standing Waves on a String, offline): learners set the fixed-end option, change frequency until the string shows a standing pattern, and count nodes and antinodes. They vary tension and record what happens to the frequency needed. They use data from the established case: $L = 0.65$ m, $v = 143$ m/s, to check the simulation setup. Each learner records at least 3 observations in a table: pattern drawn, number of nodes, number of antinodes, and effect on pitch.	each group records both slinky patterns and the length-pitch observation. Provide the term 'node' and 'antinode' only after learners describe the points in their own words.		
Explain Phase  VIDEO: Properties of periodic waves <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Standing Waves Practice Source: CK-12  Search ARES for similar readings	Learners consolidate observations into an explanation using a sequence of guided steps. Step 1: In pairs they explain how a standing wave forms: the wave travelling along the string reflects at the fixed end (linking to Lesson 3 reflection) and the reflected wave travels the opposite way; the two waves superpose (linking to Lesson 9 interference), giving fixed nodes where they always cancel and antinodes where they always reinforce. Step 2: Using the guitar string example they calculate. For the fundamental, $L = \lambda/2$ so $\lambda = 2 \times 0.65 = 1.30$ m, and from the wave equation $v = f\lambda$, $f = 143/1.30 = 110$ Hz. For the second harmonic, $\lambda = 0.65$ m (3 nodes, 2 antinodes) and $f = 143/0.65 = 220$ Hz. When the string is pressed to 0.325 m at the same tension, $\lambda = 2 \times 0.325 = 0.65$ m and $f = 220$ Hz, a higher note from the shorter string. Step 3: They explain that greater tension raises the wave speed on the string, so for the same length the frequency rises. Step 4 (link to phenomenon): The distance	Say: "Use your tables as evidence. Where do the two waves meet, and why does the pattern stay in place?" Draw a large diagram of the string with an incident wave and a reflected wave. Ask: "What happens at the fixed end to the reflected wave? Which lesson showed us reflection?" WAIT TIME 10-15 seconds. Cold-call 3 learners for the formation of the standing wave and 2 learners for each calculation step. Model the first calculation on the board: "L is half a wavelength, so λ equals 2 times 0.65." Then let learners do the second harmonic and the 0.325 m case independently, checking with a neighbour. Ask: "Why does the 0.325 m string give the same frequency as the second harmonic of the long string?" Then ask: "If nodes are half a wavelength apart and the radio wavelength is 3.0 m, how far apart are the weak spots?" Wait 10-15 seconds and cold-call 2 learners. Do not re-teach interference or reflection; refer back only: "In Lesson 9 we used	Claim-Evidence-Reasoning (CER) with mathematical modelling. Learners state a claim (shorter string, higher note), support it with observation and calculation, and give reasoning using superposition and $L = \lambda/2$. This builds a causal chain from reflection to superposition to standing pattern to note, then transfers the half-wavelength idea to the radio fading spots.	Collect pair-share written explanations. Success indicators: learners correctly state that two waves travelling in opposite directions superpose; they correctly compute 1.30 m and 110 Hz; they compute 220 Hz for both the second harmonic and the 0.325 m string; they state weak FM spots are 1.5 m apart as half of 3.0 m. Watch for errors such as using $L = \lambda$ instead of $L = \lambda/2$, or confusing nodes with antinodes. Do a quick thumbs-up/sideways check on confidence after the calculations.

	between neighbouring nodes is $\lambda/2$. The FM signal has wavelength 3.0 m, so weak spots are half a wavelength apart, which is 1.5 m, exactly the spacing noted near the Ngong Hills. Learners write a four-sentence explanation, then pair-share and revise it.	path difference; in Lesson 3 we saw reflection." Address the misconception that a standing wave does not move energy along the string; say that nodes stay still while antinodes oscillate in place.		
Driving Question Board (DQB) Creation  VIDEO: Properties of periodic waves - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Wave on a String Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners return to the class Driving Question Board with the question: 'How can we hear a matatu around a corner that we cannot see, and why does its pitch change and its radio signal fade as it moves?' In groups they write on sticky notes (a) new evidence from today, (b) a question that remains. Examples they may post: 'Standing wave nodes are half a wavelength apart, like the weak FM spots 1.5 m apart'; 'A shorter string gives a higher note'; 'Why does the matatu horn sound higher as it comes closer?' They place notes under Part A (corner), Part B (pitch), or Part C (signal fading). Learners also tick off earlier questions that now have answers and mark those that still need evidence, especially the pitch question for Part B.	Say: "Go back to our driving question. What can we now explain better than before?" Hand out sticky notes and give 3 minutes. WAIT TIME 10-15 seconds after asking: "Which part of our puzzle did today's evidence help most?" Cold-call 3 learners to read a note aloud and explain where they placed it. Ask: "Which question is still unanswered?" Highlight that the pitch change (Part B) has no explanation yet and tell learners it is the focus of the next lesson. Sort notes together and star questions that are most frequently asked. Avoid giving the Doppler explanation now; say: "Hold that question for Lesson 11."	Driving Question Board as a public record of progress. Learners connect new evidence to the anchoring phenomenon and see how knowledge accumulates, which maintains purpose and motivates the next investigation.	Count the number of sticky notes that connect standing waves to Part C (fading signal). Check that at least half the groups post a question about the matatu pitch. Look for notes that correctly state the 1.5 m link or the string-length result. Identify learners whose questions show misconceptions, for example believing standing waves are not caused by reflection.
Model Building Phase  VIDEO: Wave properties <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML:	Learners revise the class model, which has three parts: A (corner/diffraction), B (pitch/Doppler), C (signal fading/interference). Today the focus is Part C. In groups they add a diagram showing the direct FM signal (wavelength 3.0 m) and the signal reflected from the Ngong Hills hillside travelling in opposite directions, forming a standing-wave-like pattern with weak spots (nodes) 1.5 m apart and strong spots between them. They label wavelength 3.0 m, spacing $\lambda/2$ =	Say: "Add today's evidence to Part C. Show how the string helps us understand the radio fading." Remind them: "The reflected signal comes back and overlaps with the direct signal, just like the reflected wave on the string." Give 6 minutes for revision and WAIT TIME of 10-15 seconds before asking groups to explain their diagram. Cold-call 3 groups to present one change to the model and the evidence behind it. Ask: "What is the same between the guitar string and the FM signal?	Collaborative model revision with evidence-based justification. Learners decide what changes in the model and defend each change with data, which strengthens causal reasoning and shows how a single model grows across lessons.	Check that each group's Part C diagram includes: wavelength 3.0 m, weak spots 1.5 m apart, link to $\lambda/2$, and the idea of direct and reflected signals overlapping. Check that the guitar string values (0.65 m, 143 m/s, 110 Hz, 220 Hz, 0.325 m) are shown correctly. Listen for correct use of the terms node and antinode. Look for at least one group that identifies the limits of the analogy.

Standing Waves (Flexbook) Source: CK-12 Search ARES for similar readings	1.5 m, and add a mini panel showing the guitar string analogy: $L = \lambda/2$, 110 Hz, 220 Hz, 0.325 m. In Part A and Part B they add a note on what remains unchanged and what is still to be explained (Part B is still incomplete until Lesson 11). Each group adds one 'we still wonder' statement. Groups share their updated Part C diagram in a short gallery walk.	What is different?" Note that the radio case is similar because of overlapping direct and reflected signals, but do not claim a perfect standing wave in the open air. Prompt learners to leave Part B unfinished and flag what evidence is needed. Close with: "Next lesson we ask why the horn sounded 418 Hz while approaching and 383 Hz while moving away."		
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D. TEACHER REFLECTION

Which learners could explain why the pattern stays fixed, in terms of two waves moving in opposite directions, and which still described it as one wave moving? Did learners confuse $L = \lambda/2$ with $L = \lambda$, and how did the sketches reveal this? How well did learners connect the 1.5 m spacing of weak FM spots to half of the 3.0 m wavelength? Which station (slinky, guitar, simulation) gave the strongest evidence, and what could be improved if equipment was limited? Did learners notice that the shorter string gives a higher note without being prompted? What needs to be revisited before starting the Doppler effect in Lesson 11?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a slinky and a guitar string of length 0.65 m, when shaken or plucked, showed fixed points of no movement (nodes) and points of large movement (antinodes). The fundamental pattern had 2 nodes and 1 antinode; the second pattern had 3 nodes and 2 antinodes. A shorter string (0.325 m) and a tighter string gave a higher note.
What did I learn?	We learned that a standing wave forms when a wave and its reflection travel in opposite directions and superpose. Adjacent nodes are $\lambda/2$ apart, and for the fundamental $L = \lambda/2$. With $v = 143$ m/s on a 0.65 m string, $\lambda = 1.30$ m and $f = 110$ Hz; the second harmonic is 220 Hz, and a 0.325 m string also gives 220 Hz. Length and tension set the note.
How does this explain the phenomenon?	We explained that the weak FM spots near the Ngong Hills are about 1.5 m apart because the direct and reflected signals overlap, and the 100 MHz signal has wavelength 3.0 m, so weak spots are half a wavelength (1.5 m) apart, just as nodes on a string are $\lambda/2$ apart. This strengthens Part C of our model, while the pitch change of the matatu horn (Part B) still needs explaining.

LESSON 11 (40 minutes (double lesson: 80 minutes recommended if the moving-buzzer activity is run twice)): The Doppler Effect: Why the Matatu Horn Changes Pitch

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners explain and calculate why the pitch of a moving sound source seems higher on approach and lower on recession, resolving Puzzle 2 of the matatu mystery (the horn heard at 418 Hz approaching and 383 Hz receding although Baraka keeps it at 400 Hz).
Knowledge	The Doppler effect is the apparent change in the observed frequency of a wave because the source and observer move relative to each other. A moving source compresses wavelengths ahead of it and stretches them behind it. Approaching: $f' = f v/(v - v_s)$. Receding: $f' = f v/(v + v_s)$. For the matatu ($f = 400$ Hz, $v_s = 15$ m/s, $v = 340$ m/s) $f' = 418$ Hz approaching and 383 Hz receding, a 35 Hz drop. For the ambulance siren (700 Hz at 20 m/s) $f' = 744$ Hz approaching and 661 Hz receding. The horn itself never changes.
Skills	Predicting from prior experience; observing and recording pitch changes; using a model (moving buzzer, wavefront drawings) to explain observations; analysing and interpreting data; using mathematical thinking to apply the Doppler formulas and $v = f\lambda$; constructing explanations; communicating using evidence and reasoning.
Attitudes	Curiosity about everyday sounds on Nairobi roads; respect for peers' ideas during discussion; responsibility and teamwork in the moving-buzzer activity; appreciation of how physics explains daily life (ambulances, matatus, speed checks).
Key Inquiry Question	How do waves behave when they meet obstacles or different media? How are the properties of waves applied in day-to-day life?
Purpose in Storyline	Lessons 7 and 8 resolved Puzzle 1 (diffraction at the 4.0 m opening) and Lesson 9 began resolving Puzzle 3 (interference). Lesson 11 resolves Puzzle 2 (pitch change) by building on Lesson 6 ($v = f\lambda$, wavelength and frequency) and Lesson 1 pitch observations (418 Hz and 383 Hz heard, values only observed then). Learners now calculate those values and complete Part B of the class model, ready for the final synthesis in Lesson 12.
Safety Notes	Swing the buzzer on a string in a clear space, with a string no longer than 1 m, a small light buzzer, and learners standing at least 3 m from the swinging path. The swinger wears eye protection if available and keeps the swing at a steady, gentle speed. Do not swing overhead. Keep the volume of videos moderate. Walk, do not run, during the walking demonstration. Keep bags and desks clear of the activity area.

B. LESSON OVERVIEW

Learners revisit the three puzzles on the Driving Question Board and focus on Puzzle 2: why Baraka's horn sounds higher as the matatu approaches and lower as it moves away although he never changes it. After predicting, learners observe a YouTube demonstration of a moving sound source (ambulance/car horn passing), a classroom activity with a small buzzer swung on a string, and a Khan Academy explanation of the Doppler effect. They draw wavefronts for a moving source, link them to $v = f\lambda$ from Lesson 6, and use $f' = f v/(v - v_s)$ and $f' = f v/(v + v_s)$ to calculate 418 Hz and 383 Hz for the matatu (35 Hz drop) and 744 Hz and 661 Hz for the ambulance. They add new questions to the DQB and complete Part B of the class model, noting that the same idea does not yet explain the fading radio, which belongs to Part C.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Doppler effect introduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Doppler Effect (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the matatu phenomenon: a horn that Baraka keeps at 400 Hz sounds higher as the matatu approaches and suddenly lower as it moves away (418 Hz then 383 Hz, observed in Lesson 1). Individually, in their notebooks, learners write a prediction answering: 'What is happening to the sound waves that makes the pitch change, when the horn itself does not change?' They draw a quick sketch of the waves in front of and behind the moving matatu. Then they discuss in pairs: 'If you were sitting inside the matatu, would you hear the horn change pitch? Why or why not?' and 'Would a stationary horn and a walking listener give the same effect?'	Display the DQB and point to Puzzle 2. Say: 'Baraka never touches the horn. It stays at 400 Hz the whole trip. Yet Wanjiru and Otieno hear 418 Hz coming and 383 Hz going. Something is changing, and it is not the horn.' Give 2 minutes of silent writing, with WAIT TIME of 10-15 seconds after the question before anyone speaks. Circulate and note 3 common ideas (e.g. 'the horn gets louder so pitch rises', 'the matatu speeds up', 'the air pushes the sound'). Cold-call 4 learners to share predictions without judging: 'Thank you, Achieng. Who can add to or challenge that idea?' Record all predictions on the board under 'Our predictions'. Ask: 'Which prediction could we test today?' Separate loudness from pitch: 'Loudness is amplitude; pitch is related to frequency. Which one changed?'	Predict-and-sketch with Public Prediction Record: externalising initial ideas (including loudness-versus-pitch misconceptions) as sketches creates a shared baseline. Learners later compare their sketches with evidence, supporting conceptual change and ownership of the explanation.	Look for: (1) learners who confuse loudness with pitch; (2) learners who think the horn's frequency itself changes; (3) sketches that show waves squeezed together in front or stretched behind (a strong preconception). Tally on a clipboard how many of the 4 cold-called learners mention the motion of the matatu as the cause. Success indicator: at least half the class connects the pitch change to the matatu's motion, even without a mechanism.
Observe Phase  VIDEO: Doppler effect formula for observed frequency Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Doppler Effect Practice Source: CK-12  Search ARES for similar readings	Learners gather evidence in three parts. (1) Watch a short YouTube demonstration of a car or ambulance with a siren passing a stationary observer, listening for pitch on approach and recession, and a clip that shows wavefronts bunched ahead and spread behind. In a table they record: 'Moving toward me / passing / moving away' and the pitch heard (higher, same, lower). (2) Moving-buzzer activity in groups of 5: one learner swings a small buzzer on a 1 m string in a horizontal circle at steady speed in a clear space while 3 observers stand at different positions around the circle (about 3 m away) and one recorder notes whether the pitch is higher, lower or steady at each position. They repeat, then try a faster swing and	Before the video say: 'Do not just listen. Ask yourself: where is the sound higher, and where is it lower?' Play the clip twice: first for listening, second while pausing at 'approaching', 'passing' and 'receding' and asking for a show of hands for higher, same or lower. For the buzzer, demonstrate safe swinging first and state safety rules: 'Clear space, 3 metres back, gentle steady swing, no overhead swinging.' Circulate and ask each group: 'Where were you standing when the pitch was highest? What was the buzzer doing then?' After 10 minutes ask groups to report: 'Cold-call 3 groups: what did observers hear when the buzzer moved toward them? Away from them?' Prompt for the faster swing: 'What happened to the difference	Multi-source evidence gathering with a Data Table: combining video, hands-on modelling and an authoritative reading lets learners cross-check patterns (higher approaching, lower receding, bigger effect at higher speed). The physical model (the buzzer) makes the abstract wavefront model concrete, and the table organises observations for later explanation.	Observe whether each group's recorder correctly identifies higher pitch when the buzzer moves toward an observer and lower when it moves away. Check that learners notice the pitch is steady or neutral at the sideways position, and that the effect increases with speed. Ask 2 learners to explain in one sentence how the wavefront drawing shows compression ahead. Success indicators: correct table entries; consistent use of the word 'pitch' not 'loudness'; correct labelling of vs as the speed of the source.

	note whether the difference becomes bigger. (3) Khan Academy: learners read or watch the Doppler effect section from 'Waves Applications (radio, radar, Doppler)' and note the two key formulas and what v_s means. Learners then use the matatu data: with $f = 400$ Hz, $v = 340$ m/s, $v_s = 15$ m/s, they calculate wavelength ahead of the matatu and behind it.	when the speed increased?' Introduce the Khan Academy materials: 'Find the two formulas and tell your partner what v_s stands for.' Use WAIT TIME of 10-15 seconds before taking answers. Draw wavefronts for a stationary horn (circles centred on the source) and then for a moving horn (circles whose centres shift forward), labelling compressed wavefronts ahead and stretched wavefronts behind.		
Explain Phase  VIDEO: Doppler effect formula for observed frequency - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Doppler Effect (Flexbook) Source: CK-12  Search ARES for similar readings	Learners build a causal explanation of the pitch change in three steps. Step 1 (conceptual): in pairs they use a wavefront diagram to explain that each wave crest is emitted from a position closer to the observer when the source approaches, so crests arrive closer together (shorter wavelength, higher frequency); behind the source crests are emitted from farther away, so they are spaced farther apart (longer wavelength, lower frequency). Step 2 (quantitative, linking to Lesson 6): the horn's true frequency is 400 Hz so the stationary wavelength is $340/400 = 0.85$ m. In one period $T = 1/400$ s the matatu moves $15 \text{ m/s} \times (1/400) \text{ s} = 0.0375$ m. Ahead, the wavelength is $0.85 - 0.0375 = 0.8125$ m so $f' = v/\lambda' = 340/0.8125 = 418$ Hz; the Doppler formula $f' = f v/(v - v_s)$ gives $400 \times 340/325 = 418$ Hz. Behind, $\lambda' = 0.85 + 0.0375 = 0.8875$ m so $f' = 383$ Hz; formula $f' = f v/(v + v_s) = 400 \times 340/355 = 383$ Hz. The drop is $418 - 383 = 35$ Hz. Step 3 (practice): learners calculate the ambulance siren (700 Hz at 20 m/s): approaching $700 \times 340/320 = 744$ Hz; receding $700 \times 340/360 = 661$ Hz. They write a	Say: 'Wavelength is the distance between crests. If the source chases its own waves, what happens to that distance?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners to explain compression ahead and stretching behind. Model the wavefront diagram on the board step by step, then ask learners to complete the Step 2 working in pairs, circulating to check use of $v = f\lambda$ from Lesson 6. Emphasise: 'The horn is always 400 Hz. The observer receives crests at a different rate.' Introduce the formulas and ask: 'Which formula is for approaching? Look at the denominator: v minus v_s gives a smaller number, so the frequency is bigger.' Address misconceptions explicitly: 'The pitch change does not mean the horn changed, and it is not caused by loudness.' Ask learners to compute the ambulance values, then cold-call 4 learners for answers (744 Hz, 661 Hz) and ask: 'Why is the drop larger for the ambulance than for the matatu?' (higher speed and higher frequency). Ask learners to apply to Kenya: 'Where else do you notice this on Nairobi roads or on a train near Mombasa?' Finally,	Claim-Evidence-Reasoning with Mathematical Modelling: learners use the wavefront model to give a causal mechanism, then verify it quantitatively with $v = f\lambda$ and the Doppler formulas. Calculating the same values already heard (418 Hz and 383 Hz) links the model to the real phenomenon and shows it has predictive power. Comparing matatu and ambulance numbers strengthens generalisation.	Check that learners (1) correctly state that observed frequency, not source frequency, changes; (2) correctly select $f v/(v - v_s)$ for approach and $f v/(v + v_s)$ for recession; (3) arrive at 418 Hz, 383 Hz, 35 Hz, 744 Hz and 661 Hz; (4) rounded to the nearest Hz. Use a 2-question exit check on mini-whiteboards: 'Approaching or receding gives the higher pitch?' and 'What is the matatu's pitch drop?' Indicators of misunderstanding: using $v + v_s$ for approach, saying the horn changed, or confusing loudness with pitch. Re-teach with the wavefront drawing if more than a third of the class errs.

	claim-evidence-reasoning paragraph for Puzzle 2 and discuss: 'Why does the pitch change suddenly as the matatu passes, and why does it not change while it is parked?'	check a stationary-source, moving-observer extension only conceptually: 'If you walk toward a parked matatu, the crests reach you more often, so you also hear a higher pitch.' Do not go beyond the two source-moving formulas.		
Driving Question Board (DQB) Creation  VIDEO: Doppler effect formula when source is moving away Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Doppler Effect Practice Source: CK-12  Search ARES for similar readings	Learners revisit the Driving Question Board. They (1) place a green sticky note on Puzzle 2 stating what they now know ('Horn stays 400 Hz; matatu approaching gives 418 Hz and receding 383 Hz because waves are squeezed ahead and stretched behind'), (2) mark Puzzle 1 (corner) and Puzzle 3 (fading radio) as answered or partly answered based on earlier lessons (diffraction, interference), and (3) write new questions on yellow notes, for example: 'Does the Doppler effect work for light or radio?', 'Why does the pitch change smoothly or suddenly?', 'How do police radar guns use this?', 'Could a faster matatu give a bigger drop?', 'Does the radio in the matatu also fade because of the Doppler effect?' They sort the questions into 'can answer now' and 'need more investigation'.	Say: 'Go to the DQB. Which of our questions did today's evidence answer? Which new questions did it raise?' Provide WAIT TIME of 10-15 seconds, then ask 5 learners (cold-call) to read their sticky notes and justify where to place them. Prompt: 'Is the fading radio signal caused by the matatu's motion, or by the Ngong Hills reflecting the signal? What evidence do we have?' (linking to Lesson 9 interference, without re-teaching it). Highlight the question on higher speeds: 'Use the formula to predict: what would the matatu pitch be at 30 m/s?' as a take-home challenge. Group related questions (e.g. applications of Doppler: speed guns, ambulances, medical blood-flow) for Lesson 12. Praise precise wording and evidence-based reasoning.	Driving Question Board as a Tracking Tool: moving questions from 'unknown' to 'answered' and generating new ones makes progress visible and builds coherence across the unit. Sorting questions encourages learners to distinguish what current evidence can explain from what needs more investigation.	Check that notes on Puzzle 2 include both mechanism (compression and stretching of waves) and numbers (418 Hz, 383 Hz). Check that new questions are testable and linked to the phenomenon. Success indicator: at least 5 new relevant questions, and at least 3 learners explicitly state that the radio fading is not due to the Doppler effect but to interference. Record learners who still confuse the two for follow-up.
Model Building Phase  VIDEO: Doppler effect formula for observed frequency Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Position Relative	Learners revise the single class model for the matatu mystery. Part A (corner/diffraction) and Part C (signal fading/interference) are reviewed from earlier lessons. The focus now is Part B (pitch/Doppler). In groups of 4 they draw a labelled diagram of the matatu moving at 15 m/s with wavefronts bunched ahead (wavelength about 0.81 m, observed pitch 418 Hz for Otieno standing ahead) and stretched behind (wavelength about 0.89 m, observed pitch 383 Hz for Wanjiru	Display the class model poster with Parts A, B and C. Say: 'Part B is blank. Using today's evidence, what must go in it?' Give WAIT TIME of 10-15 seconds, then circulate and ask each group: 'Where is the pitch higher, and what is happening to the wavelength there? What stays the same?' Cold-call 3 groups to present, and ask the class to critique: 'Does this diagram explain why the pitch changes suddenly as the matatu passes? What evidence supports it?' Agree on	Cumulative Model Revision with Peer Critique: learners revise a shared explanatory model using new evidence, and peer critique pushes them to justify each element. Linking three parts to the same phenomenon builds coherence and prepares for the final synthesis in Lesson 12.	Use a checklist: model shows compressed wavefronts ahead and stretched behind; labels 400 Hz source, 418 Hz and 383 Hz observed; states the formulas correctly; clearly says the horn does not change; separates the Doppler idea from diffraction and interference. Success indicator: at least 80 percent of groups include all five elements. Ask a final cold-call (3 learners): 'In one sentence, why does Wanjiru hear 383 Hz when the matatu drives away?'

to Motion (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	behind), with a note that the horn's true frequency stays 400 Hz and the stationary wavelength is 0.85 m. They write a short caption using the formulas $f' = f v/(v - v_s)$ and $f' = f v/(v + v_s)$ and add an ambulance inset (744 Hz approaching, 661 Hz receding). Each group adds a 'Still to explain' box noting that Part C needs a final check and that Lesson 12 will link all three parts.	one class version and fix it to the poster. Ask: 'Which parts of the model use $v = f\lambda$? Which use ideas from earlier lessons?' Point out connections: 'Part A used wavelength compared with the 4.0 m opening; Part B uses wavelength changing because the source moves; Part C uses overlapping waves. Different mechanisms, same wave ideas.' Conclude by assigning the summary table.		
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D. TEACHER REFLECTION

Did learners clearly separate pitch (frequency) from loudness (amplitude), and did they accept that the horn's true frequency stays at 400 Hz? Which evidence (video, moving buzzer, or Khan Academy reading) most convinced them? Where did the wavefront diagram or the choice between $v - v_s$ and $v + v_s$ cause confusion, and did the 418 Hz and 383 Hz calculations make sense of what they observed in Lesson 1? Did any learners link the radio fading to the Doppler effect, and how well did they use earlier evidence on interference to correct this? What adjustments to the buzzer activity (speed, spacing of observers, safety) are needed? Which learners need additional support with substituting into the formulas, and how will the model be strengthened before Lesson 12?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the pitch of a moving horn or siren sounds higher when the source moves toward us and lower when it moves away, and that the effect is bigger for faster sources, even though the source itself does not change (matatu horn 400 Hz heard as 418 Hz and 383 Hz; ambulance 700 Hz heard as 744 Hz and 661 Hz).
What did I learn?	We learned that the Doppler effect is the apparent change in frequency because of relative motion between source and observer: approaching $f' = f v/(v - v_s)$ and receding $f' = f v/(v + v_s)$, with $v = 340$ m/s for sound and wavelength and frequency linked by the wave equation $v = f\lambda$.
How does this explain the phenomenon?	We explained that Baraka's horn stays at 400 Hz, but the moving matatu (15 m/s) compresses the waves ahead of it so Otieno hears 418 Hz, and stretches them behind it so Wanjiru hears 383 Hz, a drop of 35 Hz. This completes Part B of our class model; the corner puzzle (Part A) and the fading radio (Part C) have different causes.

LESSON 12 (2 lessons of 40 minutes (80 minutes)): Applications and Synthesis: FM, Ultrasound and the Final Explanation of the Matatu Phenomenon

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To apply wave properties to two technologies (FM radio and ultrasound imaging) and to combine evidence from Lessons 1 to 11 into one complete written and oral explanation of the matatu phenomenon.
Knowledge	Learners explain that FM radio carries sound by changing the frequency of a 100 MHz carrier wave (a 1 kHz tone changes it by up to 75 kHz) while the amplitude stays constant. They calculate the 3.0 m wavelength of the 100 MHz carrier using the wave equation $v = f\lambda$. They explain that ultrasound imaging uses reflection and echo timing (as in sonar, Lesson 4) with 3.5 MHz waves travelling at 1540 m/s, giving a wavelength of 0.44 mm. They state the final explanation: the music is heard around the corner because sound wavelengths of 0.34 m to 1.7 m are comparable to the 4.0 m opening and diffract, while light of 5.0×10^{-7} m hardly diffracts. The horn sounds 418 Hz approaching and 383 Hz receding, although it stays at 400 Hz (Doppler effect). The FM signal fades about 1.5 m apart near the Ngong Hills because direct and reflected signals interfere destructively.
Skills	Developing and using models (revise the class model Parts A, B and C); constructing explanations (claim, evidence, reasoning); analysing and interpreting data (wavelength 0.85 m, ratio 0.21, 418 Hz, 383 Hz, 1.5 m); using mathematics ($v = f\lambda$, $d = vt/2$); obtaining, evaluating and communicating information (oral and written presentation); asking questions (closing the DQB).
Attitudes	Appreciate how physics of waves is used in Kenyan communication and healthcare; show curiosity and intellectual honesty by revising earlier ideas with evidence; respect others' ideas during peer critique; take responsibility for group roles; show teamwork when presenting.
Key Inquiry Question	1. How do waves behave when they meet obstacles or different media? 2. How are the properties of waves applied in day-to-day life?
Purpose in Storyline	This is the culminating lesson. Learners apply wave properties to FM radio and ultrasound, compare their Lesson 1 model with the final class model (Parts A, B, C), close out the Driving Question Board, and deliver a complete written and oral explanation of how we hear a matatu around a corner we cannot see, why its pitch changes, and why its radio signal fades.
Safety Notes	Use only low-volume sound sources in the classroom (buzzer, speakers at safe level; no loud horn). Keep cables and speakers off walkways. Ultrasound is shown by simulation or video only; no scanner is used on learners. Move chairs carefully during gallery walk and presentations. Keep the room open so everyone can hear peers speaking.

B. LESSON OVERVIEW

Learners begin by predicting how a radio can carry a voice on a 100 MHz wave and how a doctor can see a baby without cutting. They then observe two technology examples (FM modulation using a simulation or teacher demonstration, and an ultrasound imaging video/simulation) and use $v = f\lambda$ and $d = vt/2$ to analyse them. In the Explain phase they use claim, evidence and reasoning to write a complete explanation of the matatu phenomenon, then rehearse and give an oral presentation. They close the DQB by sorting questions answered by evidence from Lessons 1 to 11, and finally compare the Lesson 1 initial model with the final class model (Part A corner/diffraction, Part B pitch/Doppler, Part C signal fading/interference). Mr Kamau, Wanjiru, Otieno and Baraka the matatu driver provide the continuing story.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Properties of periodic waves Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Interference Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Mr Kamau starts with the anchoring story: Baraka drives the Nairobi-Ngong route; the horn stays at 400 Hz and the radio at 100 MHz, yet Wanjiru and Otieno hear music around the corner, hear the pitch change, and the radio fades near the Ngong Hills. Learners individually write predictions in their notebooks for two new questions: (1) The FM station sends a voice on a 100 MHz wave. How could a wave of that frequency carry a 1 kHz voice or music? (2) A clinic in Kisumu uses sound with a frequency far above what we can hear to see a baby inside the mother. How could sound make a picture? Learners also draw a quick sketch with one sentence for each prediction, then share with a partner using a 'turn and talk' for 2 minutes.	Say: 'Today we use everything we have learned in eleven lessons. Before we learn anything new, I want your own ideas.' Display the two questions. Give WAIT TIME of 10-15 seconds in silence after each question before learners write. Circulate and note 3 different ideas, for example 'radio sends sound through air directly' or 'echo helps the picture'. Cold-call 4 learners: 'Wanjiru's seat, tell us what you think the 100 MHz wave does with the voice.' Ask: 'Which earlier lesson might help us with the baby scan? Think about Lake Victoria fishers.' Record 3 predictions on the board without judging. Say: 'We will test these by the end of the lesson.'	Predict-Observe-Explain with prior-knowledge activation: learners commit to an idea before evidence. Linking ultrasound to the Lesson 4 echo and sonar idea and the radio to the 100 MHz station in the phenomenon helps learners notice that their predictions are built from earlier evidence, so later comparison of predicted and observed ideas shows how their thinking changed.	Check notebooks for a written prediction plus a reason for both questions. Indicators: learner mentions echo, reflection or sonar when predicting about ultrasound; learner mentions 'changing' something about the radio wave when predicting about FM. Note how many of 4 cold-called learners connect to an earlier lesson. Tally common misconceptions (for example 'the voice is the radio wave') for use in Explain.
Observe Phase  VIDEO: Properties of periodic waves - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Wave Interference Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners work in groups of 4 with roles (recorder, speaker, calculator, materials manager). Station 1, FM modulation: using the PhET Sound simulation or the teacher's graphs, plus the Khan Academy Waves Applications resource (radio, radar, Doppler), they observe that the carrier is 100 MHz and a 1 kHz audio tone changes the carrier frequency by up to 75 kHz while the amplitude stays constant. They calculate the carrier wavelength using the wave equation $v = f\lambda$: $\lambda = 3.0 \times 10^8 / 1.0 \times 10^8 = 3.0$ m. They sketch a carrier whose waves bunch closer and spread farther apart following the tone. Station 2, ultrasound: they watch a short ultrasound imaging video and use the wave equation to find the wavelength of	Say: 'You are scientists collecting evidence for the final explanation. Record numbers with units.' At Station 1, ask: 'The audio tone is 1 kHz. Is it the 1 kHz that travels through the air as the radio wave? What changes in the carrier, and what stays the same?' Give WAIT TIME of 10-15 seconds, then cold-call 3 groups. Clarify: the carrier is 100 MHz = 1.0×10^8 Hz and its frequency changes by up to 75 kHz; the amplitude is constant. At Station 2, ask: 'Why is a wavelength of 0.44 mm useful for seeing small details?' and 'How is this like the Lake Victoria fisher's sonar?' Prompt: 'If an echo returns from a boundary, what do we need to know to find its depth?' Circulate with a checklist; correct unit errors (Hz vs MHz) on the spot. Do not	Data analysis with multiple representations (simulation graph, sketch, calculation, video): learners meet the same wave ideas in three forms, which strengthens understanding that FM and ultrasound rely on the same properties of waves (frequency, wavelength, speed, reflection) studied in earlier lessons. The station rotation gives repeated chances to connect each technology back to the matatu phenomenon.	Observe whether groups: (a) calculate 3.0 m for the FM carrier wavelength and 0.44 mm for ultrasound with correct units; (b) state that FM changes the carrier frequency while the amplitude stays constant; (c) use $d = vt/2$ correctly in the ultrasound explanation. Use a 3-item exit slip at each station: 'What changes in FM? What reflects in ultrasound? Which number would you use for depth?' Aim for at least 80% of groups with correct answers; reteach at the next station if fewer.

	3.5 MHz waves in soft tissue travelling at 1540 m/s: $\lambda = 1540 / 3.5 \times 10^6 = 4.4 \times 10^{-4} \text{ m} = 0.44 \text{ mm}$. They explain that pulses reflect from boundaries between tissues and the echo time gives depth using $d = vt/2$, just as in sonar on Lake Victoria (30 m depth). Station 3, evidence revisit: using their notebooks and class charts, they list the strongest evidence from Lessons 4, 8, 9 and 11 (0.85 m wavelength, 4.0 m opening, ratio 0.21, 1.5 m fading spacing, 418 Hz and 383 Hz). Groups rotate every 8 minutes.	re-teach diffraction or Doppler; point back: 'Remember Lesson 11: what did we find for the horn?'		
Explain Phase  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave on a String <i>Related topic: not an exact match for this lesson.</i> Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners build the Final Explanation using a Claim-Evidence-Reasoning (CER) frame, one section per puzzle. Part A (corner): Claim: sound is heard around the building because it diffracts. Evidence: sound wavelengths are 0.34 m to 1.7 m (0.85 m for 400 Hz) compared with the 4.0 m opening, ratio 0.21; light is $5.0 \times 10^{-7} \text{ m}$, so the ratio is about 1.3×10^{-7}. Reasoning: diffraction is greatest when wavelength is comparable to or bigger than the gap, so sound bends and light does not. Part B (pitch): Claim: the horn sounds 418 Hz approaching and 383 Hz receding although it stays at 400 Hz. Evidence: $f' = 400 \times 340 / (340 - 15) = 418 \text{ Hz}$ and $f' = 400 \times 340 / (340 + 15) = 383 \text{ Hz}$, a 35 Hz drop. Reasoning: the moving source compresses waves ahead and stretches them behind (Doppler effect). Part C (fading): Claim: the FM signal fades near the Ngong Hills because direct and reflected signals interfere. Evidence: 100 MHz gives $\lambda = 3.0 \text{ m}$; weak spots are half a	Say: 'A good explanation uses a claim, numbers as evidence, and the physics rule as reasoning. Do not just tell the story; explain why.' Provide the CER frame on the board. Give WAIT TIME of 10-15 seconds for individual thinking before groups write each section. Circulate and ask: 'Which number proves your claim? Which lesson gave you that rule?' and 'Why does light not bend round the corner while sound does? Use the ratio.' Cold-call 5 learners to read one reasoning sentence aloud and ask 2 others to add a missing piece. Listen for common errors: confusing amplitude with pitch, saying the horn really changes frequency, saying the station weakens. Correct: 'Does Baraka change the horn? Does the station change?' For presentations, assign each group one part to lead and ask the audience to give 'two stars and a wish' using the checklist. Allow about 2 minutes per group with 1 minute of questions. Use respectful language and make sure every member speaks.	Claim-Evidence-Reasoning with oral argumentation: writing and speaking require learners to connect each claim to measured data and a wave rule. Explaining aloud and receiving peer questions exposes gaps and builds a coherent causal story across diffraction, Doppler and interference, rather than three separate facts.	Use a rubric with 4 criteria: (1) all three claims correct; (2) uses exact figures (0.85 m, 4.0 m, 418 Hz, 383 Hz, 3.0 m, 1.5 m); (3) reasoning names diffraction, Doppler effect and destructive interference correctly; (4) clear oral delivery. Look for evidence that the learner states that the horn stays at 400 Hz and the station never changes. Collect written explanations to mark after the lesson. Target: at least 75% of learners score 3 of 4 or above; note learners needing support for follow-up.

	wavelength apart, 1.5 m. Reasoning: destructive interference where path difference = $(n+1/2)\lambda$, constructive between them; the station never changes. Learners add a closing paragraph linking one technology (FM modulation or ultrasound) to the properties used. Then each group rehearses a 2-minute oral presentation and groups present to the class, with listeners using a checklist.			
Driving Question Board (DQB) Creation  VIDEO: Properties of periodic waves Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Wave Interference (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners revisit the Driving Question Board created in Lesson 1: 'How can we hear a matatu around a corner that we cannot see, and why does its pitch change and its radio signal fade as it moves?' In groups, they take their own sticky-note questions from the board and sort them into three columns: Answered (with evidence and the lesson number), Partly answered, and New questions. For each Answered question they write the evidence in one sentence (for example, 'Why does the torch beam stop but music not? Lesson 8: ratio 0.21 against 1.3×10^{-7} '). They add at least one new question linked to the key inquiry about applications (for example, 'How does the phone network use waves like FM does?' or 'Why does a hospital choose 3.5 MHz?'). They identify the two KICD key inquiry questions and place their notes under them.	Say: 'Our board started as a wall of puzzles. Let's see which ones we can now answer, and with what evidence.' Give each group 5 minutes. Give WAIT TIME of 10-15 seconds after asking: 'Which question was hardest to answer and why?' Cold-call 4 groups to read one answered question and its evidence. Challenge: 'Which question can you answer only partly? What more would you need to measure?' Ask: 'Which question shows the waves behaving at obstacles, and which shows waves being applied in daily life?' Record new questions on a separate 'Next Steps' chart. Affirm learners whose questions have changed from 'what happened?' to 'why does it happen?'	Driving Question Board as a living record of learning: sorting questions by the evidence that answers them makes progress visible and shows learners that science questions are answered gradually from several investigations. New questions model how science continues and connects the lesson to the key inquiry questions.	Check that each group sorted at least 80% of its questions with a lesson number and an evidence sentence. Indicators: learner cites a number (for example 0.21, 418 Hz, 1.5 m) rather than just saying 'we learned it'; at least one new question per group that is testable or linked to a technology. Note questions that remain unanswered and record them for the teacher reflection.
Model Building Phase  VIDEO: Properties of periodic waves - dubbed(KY)	Learners place the Lesson 1 initial model beside the current class model. Working in the same groups, they finalise the three-part class model on a large chart. Part A (corner/diffraction): a plan view of the building and the 4.0 m	Say: 'A model is useful if it predicts and explains. Check that every arrow, label and number matches evidence.' Display the Lesson 1 model beside the new one. Ask: 'What did we draw in Lesson 1 for the corner? What does our model	Model revision and comparison (L1 vs final): placing the first and final models side by side shows the change in thinking and ensures that evidence drives revision. Peer feedback during the gallery walk tests whether the model explains	Checklist for each model: Part A includes the 4.0 m opening, 0.85 m wavelength and the light comparison; Part B includes 400 Hz, 418 Hz, 383 Hz and the compression/stretching idea; Part C includes 3.0 m, 1.5 m and the

Source: Khan Academy (Kiswahili) Search ARES for similar videos EXERCISE: Wave Interference Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	opening, with wavefronts of sound (0.85 m) spreading around the corner and a straight, blocked light beam (5.0×10^{-7} m). Part B (pitch/Doppler): the matatu moving at 15 m/s with wavefronts bunched ahead (418 Hz) and stretched behind (383 Hz), with the horn labelled 400 Hz at the source. Part C (signal fading/interference): direct and reflected 100 MHz signals from the Ngong Hills with wavelength 3.0 m and fading spots 1.5 m apart. Learners add a small 'applications' box showing FM modulation (carrier 100 MHz, 1 kHz tone, 75 kHz deviation) and ultrasound (3.5 MHz, 1540 m/s, 0.44 mm). They then complete a comparison table: 'What we thought in Lesson 1', 'What we know now', 'Evidence that changed our mind'. Groups do a gallery walk, leave one comment for each other group, and revise their own model after feedback.	now include that was missing?' Give WAIT TIME of 10-15 seconds before groups answer. Cold-call 4 groups to name one change and its evidence. During the gallery walk, give each learner 2 sticky notes: one 'strong because' and one 'what about...?' Ask: 'Can your model predict what happens to the corner if the horn were 1000 Hz? What would the wavelength be?' (0.34 m, bends less than 200 Hz at 1.7 m, using Lesson 8). Ask: 'If the matatu drove at the same speed away from you, would the horn change again?' Close by asking each group to state in one sentence the main idea of the unit. Collect charts and the comparison tables.	all three puzzles with one connected set of wave properties.	direct plus reflected signals; applications box includes FM and ultrasound figures correctly. In the comparison table, look for at least 3 specific changes backed by evidence. Observe whether learners accept and use peer feedback to revise. Record learners who can predict the effect of changing a variable (such as pitch or source speed).
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D. TEACHER REFLECTION

Which learners connected FM modulation and ultrasound to earlier lessons (Lesson 4 echo and sonar, Lesson 6 $v = f\lambda$) without prompting? Did any learners still think the horn really changes frequency, the radio station weakens near the Ngong Hills, or that the voice itself travels as the 100 MHz wave? Which part of the final explanation (A, B or C) needed the most support, and which pieces of evidence were most convincing to learners? Were oral presentations balanced so that quieter learners such as those who rarely spoke earlier also contributed? Which DQB questions remained unanswered and could become starting points for the next unit? Consider whether the timing of the station rotation allowed enough time for the written explanation and what to adjust next time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that music is heard around the building but the torch beam is blocked; the matatu horn sounds 418 Hz approaching and 383 Hz receding; the FM radio fades near the Ngong Hills about every 1.5 m; the FM carrier is 100 MHz and changes frequency with a 1 kHz tone (up to 75 kHz) while amplitude stays constant; ultrasound of 3.5 MHz in soft tissue at 1540 m/s has a wavelength of 0.44 mm and its echoes make an image.
What did I learn?	We learned that sound waves of wavelength 0.34 m to 1.7 m are comparable to the 4.0 m opening and diffract, while light of 5.0×10^{-7} m hardly does; that the Doppler effect changes the frequency heard when a source moves; that interference of direct

	and reflected signals makes weak and strong spots; and that FM radio and ultrasound imaging apply frequency, wavelength, reflection and the wave equation $v = f\lambda$.
How does this explain the phenomenon?	The music is heard around the corner because sound diffracts around the 4.0 m opening (0.85 m wavelength, ratio 0.21) but light does not. The horn really stays at 400 Hz; it sounds 418 Hz approaching and 383 Hz receding because the moving matatu compresses waves ahead and stretches them behind (Doppler effect). The FM signal fades near the Ngong Hills because the direct and reflected 3.0 m waves interfere destructively about 1.5 m apart, while the station never changes.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.